Spectral Selection Convolution and Overlapped-Centering Mamba Network for Hyperspectral Image Classification

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***Abstract*—Hyperspectral image (HSI) classification is a critical task in remote sensing. Recently, deep learning (DL) methods, particularly state space models (SSMs) like Mamba, have garnered significant attention due to their linear computational complexity in handling long sequences. However, processing the rich yet redundant spectral bands in HSIs remains challenging for these approaches, often resulting in models with excessive parameters. Moreover, existing DL-based methods frequently overlook explicit cross-channel dependencies. This limitation hinders their ability to effectively aggregate relevant information from redundant spectral bands. Additionally, the heavy reliance of DL-based models on extensive training data conflicts with the scarcity of labeled samples in HSI scenarios. This necessitates the incorporation of prior knowledge to mitigate data limitations. To address these challenges, we propose SliMamba, a lightweight Convolution-Mamba architecture for HSI classification (HSIC). SliMamba introduces two key components: spectral selection convolution (SSC) and overlapped-centering mamba (OC-Mamba). SSC swaps dimensions between spectral and spatial axes while preserving the original spectral information. It then utilizes a small set of convolutional kernels to learn spectral weights within the receptive field, thereby achieving cross-channel feature representation. OC-Mamba integrates prior knowledge of the center pixel with local feature enhancement. It combines one-pixel overlapped SSM (O-SSM) and centering SSM (C-SSM) to extract global–local and spectral–spatial features from HSI feature maps. Extensive evaluations on four benchmark datasets demonstrate that SliMamba consistently outperforms state-of-the-art methods in classification accuracy while significantly reducing model complexity. The source code is available at https://github.com/flyzzie/SliMamba**

***Index Terms*—Hyperspectral image (HSI) classification, overlapped-centering mamba (OC-Mamba), spectral selection convolution (SSC), state space model (SSM).**

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I. Introduction

HYPERSPECTRAL image (HSI), a specialized form of remote sensing imagery, captures abundant spectral information across a wide range of wavelengths, from visible to near-infrared. Unlike traditional images, HSIs provide detailed spectral signatures for each pixel, enabling precise identification and characterization of surface materials and objects [1].

HSI classification (HSIC) [2] is a cornerstone technique in remote sensing, aiming to assign accurate land cover category labels to every pixel in an HSI scene. In recent years, HSIC has been widely applied in intelligent agriculture, environmental monitoring, disaster detection, and other areas. For instance, Ahmed et al. [3] constructed a hyperspectral dataset specifically designed to distinguish between soybeans and weeds, supporting weed control in precision agriculture. Zhang et al. [4] developed two large-scale crop classification datasets with substantial and well-balanced label samples. Yang et al. [5] proposed a fine-grained mangrove species classification method based on UAV-acquired hyperspectral data, providing highly accurate spatial distribution maps that are crucial for coastal biodiversity conservation. Huang et al. [6] reconstructed hyperspectral representations from visible light images of PM2.5 to enrich spectral information and effectively classified atmospheric pollution levels using a tailored classification network. Li et al. [7] built multiple hyperspectral datasets for tree species diversity monitoring and proposed a texture-aware self-attention model. Despite its enormous potential, HSIC faces significant challenges due to the high spectral dimensionality, spectral variability, and limited labeled training data inherent in HSIs. These factors place stringent demands on classification models, necessitating both high accuracy and strong generalization capability.

Early research on HSIC primarily relied on traditional machine learning (ML) algorithms, such as support vector machines (SVMs) [8], K-means [9], random forests [10], and K-nearest neighbors [11]. These methods are heavily depended on manual feature extraction, which made them vulnerable to the "curse of dimensionality" when dealing with high-dimensional data. Additionally, traditional approaches typically focused solely on spectral information, making it difficult to effectively integrate both spectral and spatial feature.

In recent years, with advancements in deep learning (DL), a surge of DL-based HSIC models has emerged. These

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models leverage deep network architectures with powerful feature extraction capabilities, allowing for a greater focus on both spectral and spatial characteristics of HSI data. DL-based models are typically categorized by their structural design, including convolutional neural networks (CNNs) [12], [13], [14], graph convolutional networks (GCNs) [15], [16], recurrent neural networks (RNNs) [17], [18], Vision Transformers (ViTs) [19], [20], and state space models (SSMs) [21], [22], [23].

In fact, the aforementioned DL-based HSIC methods still face three major challenges: First, the scale of the DL-based models is positively correlated with the spectral dimension of HSI. Second, DL-based models rely on local aggregation of adjacent bands or global feature. They generally fail to explicitly model cross-channel dependencies, hindering the effective separation of discriminative spectral features from redundancy. Third, DL-based models require a large number of training samples to achieve satisfactory performance; however, the availability of labeled HSI data is often limited.

To address the first challenge, some researchers [24], [25], [26] have proposed dividing HSIs into nonoverlapping spectral groups and performing feature extraction solely within each group, which not only enhances the modeling of local spectral features but also significantly reduces the model size. However, fixed grouping cannot effectively distinguish between different land cover categories and lacks generalization. Other researchers attempted to address the first two challenges simultaneously. They apply unsupervised dimensionality reduction algorithms, such as principal component analysis (PCA) [21], [27], [28] or singular value decomposition (SVD) [29], to the spectral dimension before feeding the feature maps into the feature extraction network. This not only alleviates the problem of high spectral dimensionality but also facilitates the selection of spectral channels. While these methods can preserve most of the spectral features, they often disrupt the continuity of the original spectral and obscure the inherent physical meaning of the spectral information. As for the third challenge, researchers have incorporated prior knowledge into HSI feature maps to enhance network performance with small training samples, utilizing techniques such as Gabor filtering [30], [31], [32] and wavelet transforms [33], [34]. Additionally, other researchers have developed patch-level methods based on the prior knowledge of the labeled center pixel, expanding the feature representation of the center pixel. In patch-level HSIC methods, the input samples typically consist of the labeled central pixel and its neighboring unlabeled pixels, forming a patch (all along the spectral dimension). Li et al. [35] explored the differences between the central pixel and its surrounding pixels by calculating their similarity, thereby enhancing the discriminative power of the central pixel. Similarly, Jia et al. [36] proposed a pretraining approach where the central pixel is masked and reconstructed using the surrounding pixels, effectively capturing the contextual relationships between the central pixel and its neighbors. These methods leverage the prior knowledge of the central pixel to improve the model's feature extraction capability. However, the incorporation of the central pixel prior significantly increases the complexity of the model.

To resolve the issues discussed above, we propose a novel **li**ghtweight Convolution-Mamba network model. This architecture is termed the "spectral selection convolution and overlapped-centering **mamba** network" and is hereafter referred to as SliMamba. We introduce an original spectral selection convolution (SSC) technique. This method enables the network to perform spectral band selection using a minimal set of convolutional kernels, thereby enhancing the capability to capture cross-channel features in HSIs. Additionally, we optimize the Mamba framework by designing the overlapped-centering mamba (OC-Mamba) block. Comprising the one-pixel overlapped SSM (O-SSM) and centering SSM (C-SSM), this block refines spectral–spatial feature extraction by integrating local–global relationships with center pixel prior knowledge. Our contributions are summarized as follows.

1) First, we introduce an SSC. By employing depthwise convolutional kernels along the spectral dimension, the network adaptively performs large receptive field spectral cross-channel selection during the convolution process.
2) Second, we construct an OC-Mamba block, which emphasizes the prior knowledge of the center pixel. The block mainly consists of the O-SSM and the CSSM, which respectively model the local and global spectral–spatial features.
3) Third, we introduce a centering spectral–spatial cross-attention (CS^2CA) mechanism, which enhances both spectral and spatial information within the SSC. This module facilitates interaction between the center pixels of the OC-Mamba and SSC components, thereby supplementing the label features with prior knowledge.
4) Finally, we construct a lightweight network model and conduct extensive experiments on four real-world hyperspectral datasets. The experimental results demonstrate that our proposed SliMamba model achieves excellent performance with a significantly small model scale, compared to several state-of-the-art CNN, ViT, and SSM models previously applied to HSIC.

The remainder of this article is organized as follows. Section III presents the architecture of the proposed SliMamba model in a step-by-step fashion. Section IV presents comprehensive experimental assessment of the SliMamba model, which is carried out on four benchmark hyperspectral datasets. Section V concludes this article with some remarks and hints at plausible future research lines.

II. Related Work

A. HSIC Methods

CNN-based models are among the most widely used architectures in HSIC. By leveraging the strong local receptive fields of convolutional kernels, CNNs effectively capture the local spectral–spatial feature distributions. The evolution of CNN-based approaches for HSIC has seen significant advancements, progressing from 2D-CNNs [37] to 3D-CNNs [38], hybrid models [39], [40] that combine 2D-CNNs and 3D-CNNs, and architectures [41], [42], [43] incorporating residual connections. Wang et al. [44] proposed a hyperkernel convolution method integrated with neural architecture search, which substantially reduces the computational cost associated with model search. However, CNN-based models are constrained by their focus on local feature extraction and lack a global receptive field, both spectral and spatial, which has led to the development of ViT-based models.

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The Transformer blocks in ViTs [45] utilize multihead self-attention mechanisms, which are highly effective at modeling long-range dependencies. These mechanisms capture global spectral and spatial relationships across image patches, enhancing the model's ability to identify complex ground objects. Zhong et al. [46] were the first to propose replacing convolutions with spectral–spatial transformers and introduced a factorized architecture search (FAS) framework to enable stable and efficient architecture exploration. Additionally, several studies [20], [24] have combined the strengths of CNNs and ViTs, achieving notable performance improvements in HSIC. However, the multihead self-attention mechanism in ViTs comes with a computational complexity that scales quadratically with the input sequence length, resulting in substantial time costs during model training.

B. State Space Models

Recently, Mamba [47], built upon SSMs, has gained significant attention due to its ability to model long-sequence feature correlations with linear time complexity. Mamba initially demonstrated substantial potential in natural language processing (NLP) tasks involving 1-D sequence modeling.

1) State Space Models: SSMs [48], [49] can be regarded as linear time-invariant systems (LTI systems) that describe the mapping of input $x(t) \in \mathbb{R}^L$ to output $y(t) \in \mathbb{R}^L$ through a hidden state $h(t) \in \mathbb{R}^M$. The system is defined as follows:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t) \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{M\times M}$, $\mathbf{B} \in \mathbb{R}^{M\times 1}$, and $\mathbf{C} \in \mathbb{R}^{1\times M}$.

To convert from the continuous form to the discrete form, a zero-order hold (ZOH) method with a timescale Δ is applied. The continuous variables $\mathbf{A}$ and $\mathbf{B}$ are transformed into their discrete counterparts $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$, defined as follows:

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right)\cdot\mathbf{B}. \tag{2}$$

The discrete state space system is described as follows:

$$h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t, \quad y_t = \mathbf{C}h_t. \tag{3}$$

From the perspective of global convolution, the system can be rewritten as follows:

$$\bar{\mathbf{K}} = \left(\mathbf{CB}, \mathbf{CAB}, \ldots, \mathbf{CA}^{L-1}\mathbf{B}\right), \quad \mathbf{y} = \mathbf{x} * \bar{\mathbf{K}} \tag{4}$$

where $*$ denotes the convolution operation, L is the sequence length, and $\bar{\mathbf{K}} \in \mathbb{R}^L$ represents the SSM kernel.

2) Selective SSMs (S6): Traditional SSMs face certain limitations in capturing contextual relationships due to their static parameterization. To address this issue, Mamba [47] proposed the selective state space models (S6) to enable more flexible dynamic interactions between sequential states. Specifically, the parameters $\mathbf{B} \in \mathbb{R}^{B\times L\times M}$, $\mathbf{C} \in \mathbb{R}^{B\times L\times M}$, and $\Delta \in \mathbb{R}^{B\times L\times D}$ are dynamically computed from the input sequence $\mathbf{x} \in \mathbb{R}^{B\times L\times D}$. This design allows the model to achieve selective perception at each sequence unit, thereby effectively capturing richer contextual information.

Mamba quickly gained prominence in the field of computer vision, with innovations such as the multidirectional 2-D scanning mechanism [50] and the bidirectional Mamba block based on positional embeddings [51] being successfully applied to various downstream image tasks. Subsequently, Mamba gradually became popular across a range of remote sensing applications, including remote sensing image classification [52], semantic segmentation [53], [54], and change detection [55]. Li et al. [28] were the first to apply Mamba to HSIC, designing both spectral and spatial Mamba blocks to adaptively integrate spectral and spatial information. To address the inherent 3-D nature of HSIs, He et al. [22] introduced a 3-D spectral–spatial selective scanning (3DSS) mechanism, creating five scanning paths to model spectral–spatial features with varying priorities. Zhou et al. [21] proposed a novel multiscale pyramid Mamba model, which uses a cross-scan approach to serialize HSIs data. Some researchers exploit the continuous spectral characteristics of hyperspectral data by performing interval [56] or overlapping [57] grouping of spectral channels before conducting sequential feature modeling based on SSMs. Regarding center pixel priors, Liu et al. [58] conduct Mamba-based sequence modeling after computing center-based cosine similarity, while Tang et al. [59] propose a spiral scanning strategy that starts and ends at the central pixel to capture sequential features. In addition, Ahmad et al. [60] have explored SSM-based sequential modeling following frequency domain transformations, such as wavelet transforms. Although these methods have adapted Mamba for HSI tasks with satisfactory results, they often neglect local feature modeling or suffer from excessive parameter overhead.

III. Proposed SliMamba Model

This section provides a detailed exposition of our proposed SliMamba model. First, we elucidate the overall framework of the SliMamba model and its workflow for HSIC. Subsequently, we delve into the intricate design of the key components within the SliMamba architecture.

A. Framework of SliMamba Model

The proposed model architecture is illustrated in Fig. 1. The model consists of two feature extraction branches, a feature fusion module, and a classifier. The original branch is primarily designed to extract features from the raw spectral bands, while the dimensionality-reduction branch focuses on capturing features derived from spectrally reduced dimensions. The feature fusion module integrates features from the two branches, and the classifier generates the final classification results.

Specifically, before being fed into the network, the raw HSI is first divided into two branches. The original branch retains the original spectral bands, while the other applies PCA to reduce spectral dimension. Both branches are then cropped into overlapping image patches, centered on a labeled pixel and including its surrounding neighborhood pixels. These patches are subsequently fed into the feature extraction network for further processing.

Subsequently, the feature map is integrated with the center pixel vector (CPV) from the dimensionality-reduction branch using the SSC module. It is then further refined through the CS^{2}CA block within the fusion module.

For the dimensionality-reduction branch, the feature map incorporates the CPV provided by the (CS^{2}CA) module before being processed through the OC-Mamba block.

The feature fusion module combines the feature maps from both branches, along with the PCA-precessed feature map, by

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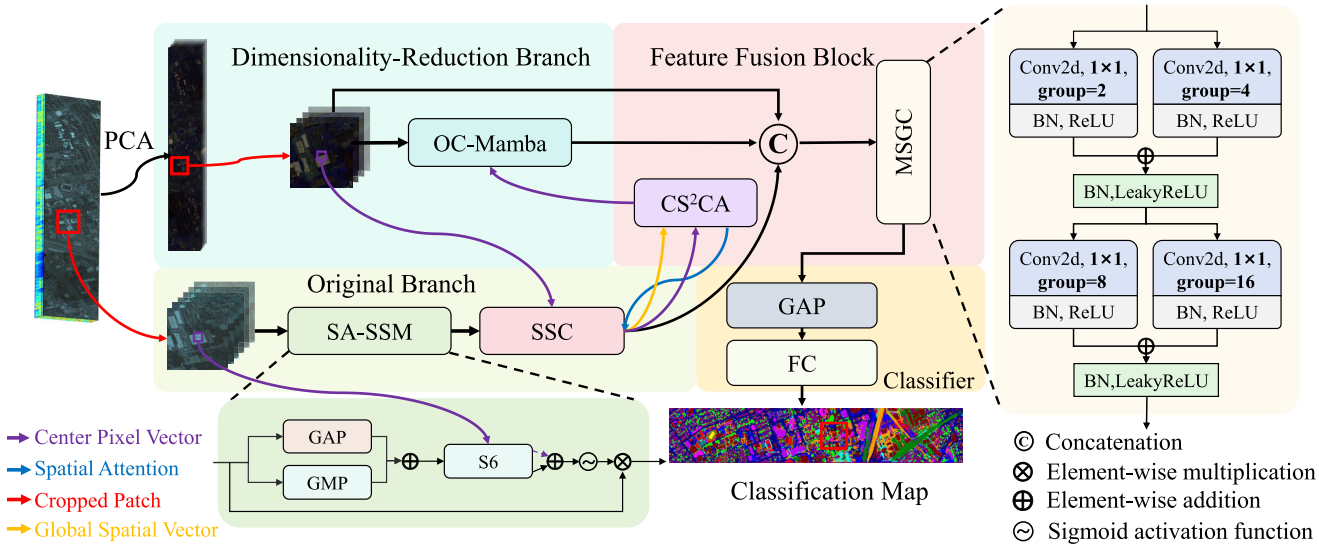


Fig. 1. Schematic of the proposed SliMamba model, in which "PCA" denotes the principal component analysis, "SA-SSM" refers to the spectral attention SSM, "SSC" refers to the spectral selection convolution, "OC-Mamba" stands for the overlapped-centering mamba, "CS^{2}CA" denotes the centering spectral–spatial cross-attention, "MSGC" denotes the multiscale groupwise convolution block, "GAP" denotes the global average pooling layer, "GMP" denotes the global max pooling layer, "FC" refers to the fully connected layer, "BN" refers to the batch normalization layer, and "ReLU" and "LeakyReLU" stand for the activate functions.

concatenating them along the channel dimension. Multiscale grouped convolution (MSGC) blocks are then employed to fuse the features.

Finally, the fused feature map is fed into the classifier, which outputs the predicted classes and generates the classification result map.

B. Architecture of SA-SSM Block

To emphasize the spectral bands of interest in the original spectrum while ignoring irrelevant bands, we propose a spectral attention SSM (SA-SSM), which leverages SSM as an attention mechanism to efficiently model the global spectral correlations.

As illustrated in the SA-SSM module in Fig. 1, for the input feature map $\mathbf{F}_a \in \mathbb{R}^{P\times P\times C}$, global average pooling (GAP) and global max pooling (GMP) are first applied separately to obtain two feature maps with a spatial size of 1, retaining only spectral features. The resulting feature maps are $\mathbf{F}_{\text{GAP}} \in \mathbb{R}^{1\times 1\times C}$ and $\mathbf{F}_{\text{GMP}} \in \mathbb{R}^{1\times 1\times C}$. These two feature maps are then added elementwise to produce $\mathbf{F}_{G(A+M)P} \in \mathbb{R}^{1\times 1\times C}$.

Next, the original CPV $\mathbf{F}_{\text{center}} \in \mathbb{R}^{1\times 1\times C}$ and the pooled feature map $\mathbf{F}_{G(A+M)P}$ are fed into the S6 module. The outputs of this module are then added together and passed through a sigmoid activation function to obtain the global spectral attention. Finally, this attention is applied to the original feature map through elementwise multiplication. The detailed process is expressed as follows:

$$\begin{aligned}\mathbf{F}_{G(A+M)P} &= \mathbf{F}_{\text{GAP}} \oplus \mathbf{F}_{\text{GMP}}\\ \mathbf{F}_{P_S6}, \mathbf{F}_{C_S6} &= S6\left(\mathbf{F}_{G(A+M)P}, \mathbf{F}_{\text{center}}\right)\\ \mathbf{F}_{\text{SA-SSM}} &= \mathbf{F}_a \otimes \text{Sigmoid}\left(\mathbf{F}_{P_S6} \oplus \mathbf{F}_{C_S6}\right)\end{aligned} \tag{5}$$

where $\mathbf{F}_{P_S6}$ and $\mathbf{F}_{C_S6}$ denote the outputs of the S6 block, and $\mathbf{F}_{\text{SA-SSM}}$ denotes the output feature map of the SA-SSM module.

C. Architecture of SSC

The feature maps from the original branch are processed by the SA-SSM module and then fed into the SSC block. To address the lack of cross-channel feature modeling in the spectral (also called channel) dimension of HSIs by conventional convolutional approaches, we propose a lightweight SSC technique.

Fig. 2(a) illustrates the detailed process of SSC, which consists of five main steps. The main stages are highlighted by blue dashed boxes, with each step's corresponding feature map shown to the left of the associated arrow. Each step is numbered with circled digits and labeled in the figure. Suppose the input feature map to SSC from the original branch is $\mathbf{F}_0 \in \mathbb{R}^{P\times P\times C}$, and the output is $\mathbf{F}_5 \in \mathbb{R}^{P\times P\times C'}$. To ensure that the center pixel of the input feature map corresponds to a single pixel, we set P as an odd number, which can be expressed as $P = 2kn + 1$, where n is a positive integer, and k is an expansion factor, also a positive integer. Detailed steps are listed below.

Step ①: First, the process begins with convolution operations applied to two branches. The first branch employs a pointwise grouped convolution with a group size of $4k$. This operation performs a coarse aggregation of local spectral features while compressing the original spectral dimension and not altering the spatial dimension. The precondition for grouped convolution is that both the input and output channels must be divisible by the group size. To meet this condition, mirrored padding (as described in [61]) is applied to ensure that the number of input feature map's channels is divisible by $4k$. Similarly, the output feature map's channel count is set to $P^2 - 1$, which is also divisible by $4k$. This divisibility is mathematically proven using the difference of squares formula

$$\begin{aligned}P^2 - 1 &= (P+1)(P-1)\\ &= (2kn+1+1)(2kn+1-1)\\ &= (2kn+2)(2kn) = 4k\cdot(kn+1)n.\end{aligned} \tag{6}$$

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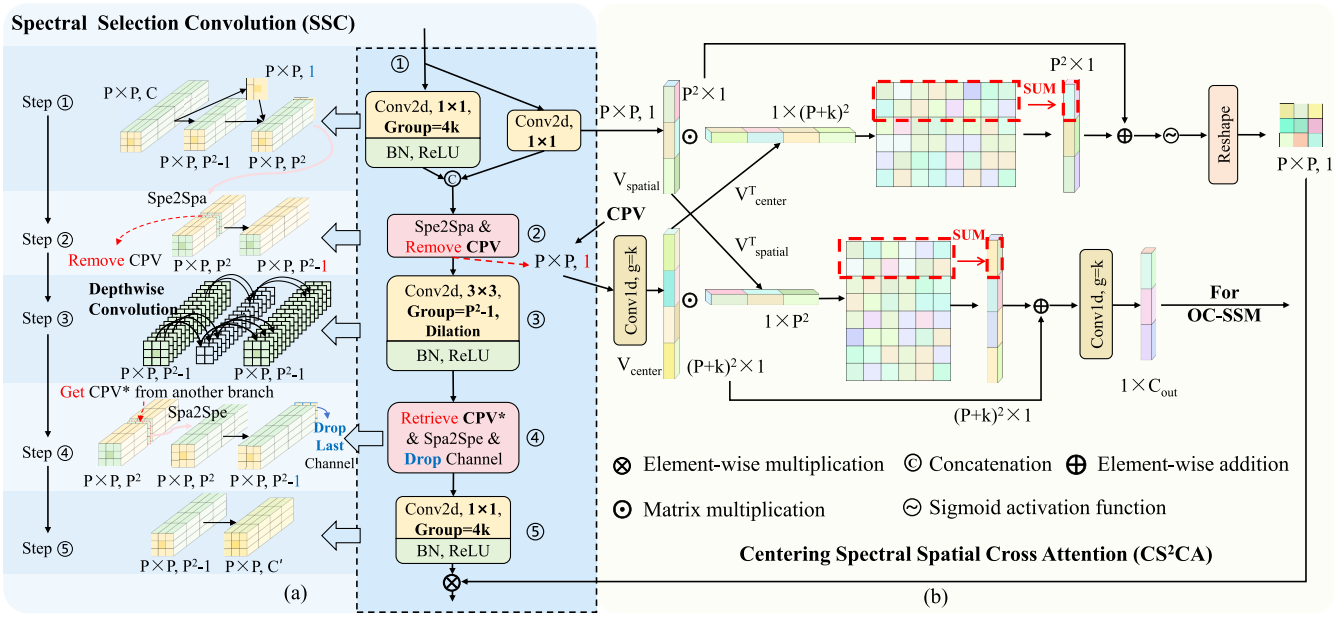


Fig. 2. (a) Overall framework of SSC, (b) CS^{2}CA process. The operations at each step and their corresponding feature maps (with specific dimensions) are illustrated in (a). The specific step numbers are indicated by circled Arabic numerals. 'BN' refers to batch normalization, and "ReLU" stands for the activation function in Steps ①, ③, and ⑤. In Step ②, "Spe2Spa" represents the transformation of the original spectral dimension into the spatial dimension (with the original spatial dimension being converted into the spectral dimension), and "CPV" refers to the center pixel vector. In Step ④, "CPV*" denotes the center pixel vector obtained from the dimensionality-reduction branch, and "Spa2Spe" represents the conversion of the current spatial dimension back into the spectral dimension (i.e., restoring the feature map to its original corresponding spectral space dimension).

The second branch applies a pointwise convolution to compress the spectral dimension of the original feature map to 1, obtaining a global spatial feature representation. The outputs of both branches are then concatenated along the channel dimension, resulting in $\mathbf{F}_1 \in \mathbb{R}^{P\times P\times P^2}$. Detailed process of the step is presented in below

$$\mathbf{F}_1 = \text{Concat}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv2d}_{1\times1,g=4k}\left(\mathbf{F}_0\right)\right)\right) \times \text{Conv2d}_{1\times1}\left(\mathbf{F}_0\right)\right) \tag{7}$$

where g refers to the number of groups during convolution and $\mathbf{F}_0 \in \mathbb{R}^{P\times P\times C}$ represents the input feature map.

Step ②: Next, the spectral and spatial dimensions are exchanged. The spatial dimensions are flattened into a single dimension to form a new spectral dimension, while the spectral dimension is expanded into a 2-D structure, becoming the new spatial dimensions. During this operation, the CPV from the original spatial dimension (now the new spectral dimension) is removed and temporarily set aside. After this transformation, the feature map becomes $\mathbf{F}_2 \in \mathbb{R}^{P\times P\times(P^2-1)}$, which can be noted as the following equation:

$$\mathbf{F}_2 = \text{RemoveCPV}\left(\text{Spe2Spa}\left(\mathbf{F}_1\right)\right) \tag{8}$$

where CPV stands for the center pixel vector of $\mathbf{F}_1$, and Spe2Spa refers to the dimensionality exchange.

Step ③: Following the dimensionality exchange, a convolution operation is applied to the transformed feature map. Specifically, a 3×3 depthwise convolution is used to extract cross-channel features in the original spectral dimension. This operation does not change the feature map's size, resulting in $\mathbf{F}_3 \in \mathbb{R}^{P\times P\times(P^2-1)}$. The equation of this step is organized below

$$\mathbf{F}_3 = \text{ReLU}\left(\text{BN}\left(\text{Conv2d}_{3\times3,g=P^2-1}\left(\mathbf{F}_2\right)\right)\right). \tag{9}$$

Step ④: The process then reintegrates the CPV. The previously extracted CPV is appended to the current spectral dimension (original spatial dimension), and the spectral and spatial dimensions are restored to their original configuration. At this stage, the feature map has dimensions $P\times P\times P^2$. Since a global spatial feature representation was concatenated in Step ①, the last channel of the current feature map is dropped to restore the feature map. This step yields $\mathbf{F}_4 \in \mathbb{R}^{P\times P\times(P^2-1)}$, where the number of channels remains divisible by $4k$. This step is presented in the following equation:

$$\mathbf{F}_4 = \text{DropChannel}\left(\text{Spa2Spe}\left(\text{RetrieveCPV}^*\left(\mathbf{F}_3\right)\right)\right) \tag{10}$$

where CPV* presents the center pixel vector from the input feature map of dimensionality-reduction branch, and Spa2Spe refers to the dimensionality restoration.

Step ⑤: Finally, a pointwise grouped convolution with a group size of $4k$ is applied to compress and integrate the local channel features, producing the output feature map $\mathbf{F}_5 \in \mathbb{R}^{P\times P\times C'}$. Below equation describes the step

$$\mathbf{F}_5 = \text{ReLU}\left(\text{BN}\left(\text{Conv2d}_{1\times1,g=4k}\left(\mathbf{F}_4\right)\right)\right). \tag{11}$$

Specifically, as illustrated in Fig. 3, we explain the SSC mechanism from the perspective of the original spectral dimension of the feature map. For clarity, Roman numerals are used in the diagram to represent each stage of the feature map. Suppose the spatial size of the input feature map I is 7×7, a group number of 4 ($k=1$), and an input channel number C (divisible by 4). Since a grouped convolution with 4 groups is applied, spectral features interact only within nonoverlapping groups. In the figure, four spectral groups are represented using approximately but distinct colors. In Step ②, the original spectral dimension has been transformed into

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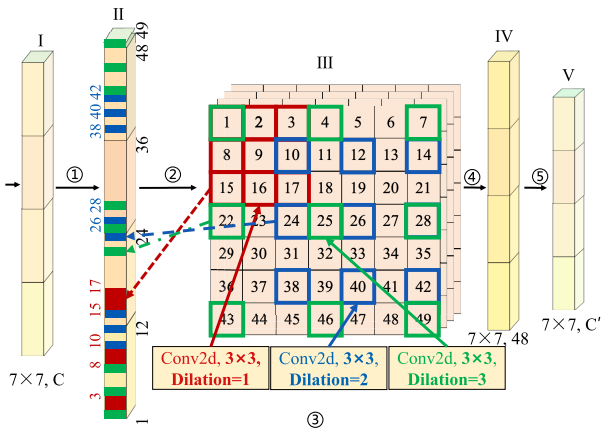


Fig. 3. Execution process of SSC from the spectral perspective. The Roman numerals denote the feature maps at different stages. In Step ③, depthwise convolutions with different dilation values are represented by different colors, applied to the feature map III, and reflected in the feature map II.

the current spatial dimension. In feature map III, we label the spatial dimensions (1–49), which correspond to the spectral dimension in feature map II.

The key operation occurs in Step ③, where a depthwise convolution operation is performed on the feature map after dimension swapping. We use red, blue, and green to represent the cases where the dilation values of the 3×3 convolution kernel are 1, 2, and 3, respectively. The depthwise convolution process involves sliding the convolution kernel over the spatial dimensions (original spectral dimensions) of a specific channel in the feature map, performing elementwise multiplication and addition, and ultimately outputting the current channel. In Step ③ of SSC, the diffusion degree of the convolution kernel determines its receptive field. As the dilation values increase, the spectral selection range of the convolution kernel transitions from being concentrated to more dispersed, spanning an increasing number of groups. This indicates a broader spectral selection range. Our SSC innovatively exchanges the spectral and spatial dimensions of the feature map, enabling depthwise convolution operations performed along the current spatial dimension to correspond to the spectral dimension prior to the transformation. By leveraging the weight adjustments of the convolution kernels during the training process, SSC effectively selects spectral bands within the receptive field, thereby facilitating the extraction of cross-channel spectral features.

Additionally, we compare SSC with the widely adopted and efficient depthwise separable convolution (DSC). The key distinctions are observed in two aspects. First, the primary difference lies in the feature extraction mechanism. SSC is specifically designed to capture cross-channel spectral interactions. In contrast, DSC primarily focuses on spatial feature extraction. Although DSC incorporates global channel interactions via pointwise convolutions, it lacks the capability to explicitly model localized spectral dependencies along the spectral axis. Second, SSC demonstrates superior parameter efficiency compared to DSC. Consider an input with a patch size of $P \times P$ (where $P = 2\text{nk} + 1$), C input channels, and C' output channels. Assuming a kernel size of 3, the parameter count for SSC is calculated as $C + (C(P^2-1)/4k) + 9(P^2-1) + (C'(P^2-1)/4k)$. Conversely, the parameter count for DSC is $9C + \text{CC}'$. Under typical experimental settings (e.g., $n = 2$, $k = 2$, $C = 128$, and $C' = 32$), SSC requires only 2448 parameters. This is less than half of the 5248 parameters required by DSC. Notably, the parameter reduction achieved by SSC becomes increasingly significant when the receptive field is small.

D. Architecture of CS²CA Block

As shown in the right diagram of Fig. 2, in the CS²CA module, we perform interactions between the global spatial vector (GSV) generated by the second branch in Step ① of SSC and the CPV extracted in Step ②. The GSV can be illustrated as $\mathbf{v}_{\text{spatial}} \in \mathbb{R}^{P^2}$. As for the CPV, in order to enhance local spectral information, we introduce a groupwise convolution for local channel interaction. We transform the CPV from $\mathbf{v} \in \mathbb{R}^{P^2}$ into $\mathbf{v}_{\text{center}} \in \mathbb{R}^{(P+k)^2}$ by applying a 1-D groupwise convolution with a group size of k.

To enhance the interaction between the spectral features of the CPV and the spatial features of the GSV, we employ cross-correlation operations to capture their relationships across multiple granular levels, as expressed below

$$\mathbf{M} = \mathbf{v}_{\text{spatial}} \cdot \mathbf{v}_{\text{center}}^{T} \tag{12}$$

where $\mathbf{M}$ denotes the correlation matrix. We adopt a straightforward approach by extracting the row and column information directly from the correlation matrix and its transpose, which are used as adjusted weight vectors for spatial and spectral features, as shown below

$$\begin{aligned}
\mathbf{v}_{\text{spatial}}^{w} &= \sum_{j=1}^{P^2} \mathbf{M}_{i,j},\ i \in 1,2,\ldots,P^2,\ j \in 1,2,\ldots,(P+k)^2 \\
\mathbf{W}_{\text{spatial}} &= \text{Reshape}\left(\text{Sigmoid}\left(\mathbf{v}_{\text{spatial}} \oplus \mathbf{v}_{\text{spatial}}^{w}\right)\right) \\
\mathbf{F}_{\text{SSC}} &= \mathbf{W}_{\text{spatial}} \otimes \mathbf{F}_5
\end{aligned} \tag{13}$$

$$\begin{aligned}
\mathbf{v}_{\text{center}}^{w} &= \sum_{j=1}^{(P+k)^2} \left(\mathbf{v}_{\text{center}} \cdot \mathbf{v}_{\text{spatial}}^{T}\right)_{i,j} \\
&= \sum_{j=1}^{(P+k)^2} \mathbf{M}_{i,j}^{T},\ i \in 1,2,\ldots,(P+k)^2 \\
&\qquad j \in 1,2,\ldots,P^2 \\
\mathbf{v}_{c} &= \text{Conv1d}\left(\mathbf{v}_{\text{center}} \oplus \mathbf{v}_{\text{center}}^{w}\right)
\end{aligned} \tag{14}$$

where $\mathbf{v}_{\text{spatial}}^{w}$ and $\mathbf{v}_{\text{center}}^{w}$ are the spatial weights vector and spectral weights vector after weight interaction, and P^2 and $(P+k)^2$ are the number of channels of $\mathbf{v}_{\text{spatial}}^{w}$ and $\mathbf{v}_{\text{center}}^{w}$, respectively.

To preserve the original features, we add the weight-interacted vector $\mathbf{v}_{\text{spatial}}^{w}$ back to the preinteraction vector $\mathbf{v}_{\text{spatial}}$, similar to the operation in CPV. For GSV, normalization is performed using the sigmoid activation function. As shown in (13), subsequently, the resulting matrix is reshaped to $\mathbf{W}_{\text{spatial}} \in \mathbb{R}^{P \times P \times 1}$ and multiplied elementwise with the feature map $\mathbf{F}_5$ from the SSC output to supplement the global spatial features. For the CPV, we further apply a \text {Conv1d} operation with a group number of k to transform it into the same number of channels as the dimensionality-reduction branch feature map, facilitating subsequent operations, as shown in (14).

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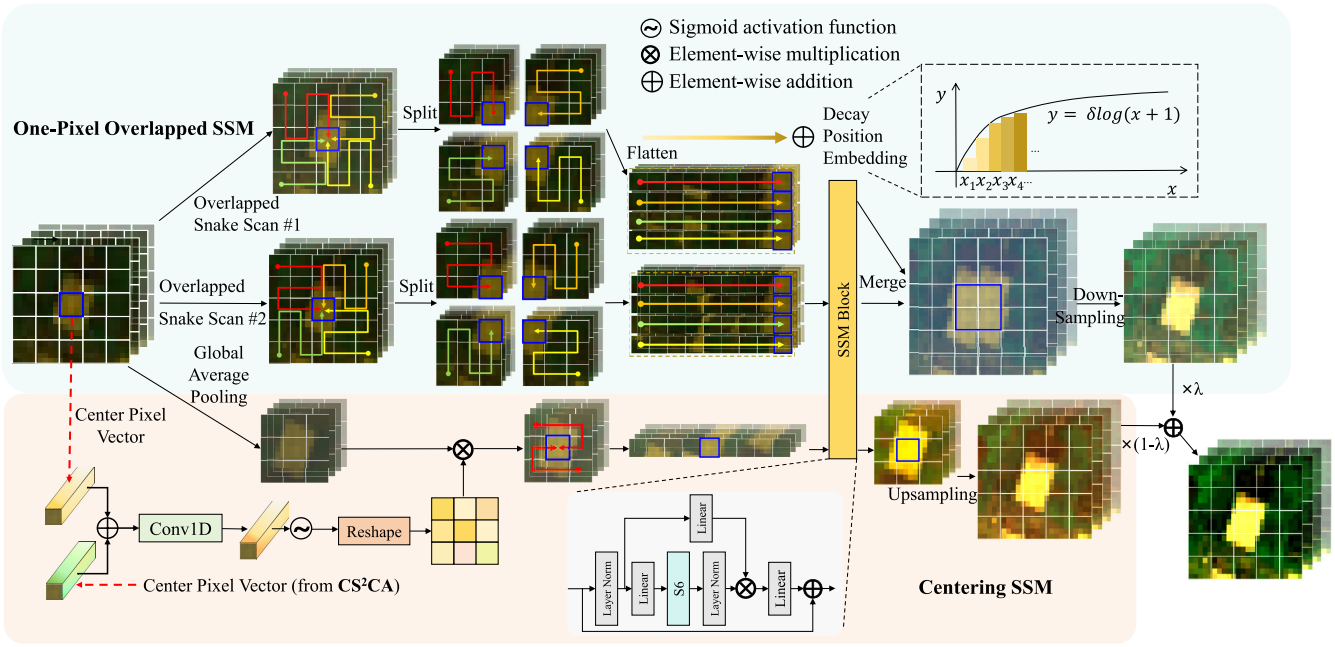


Fig. 4. Framework of OC-Mamba.

E. Architecture of OC-Mamba Block

To address the limitation of existing SSM-based models in integrating local and global spectral–spatial feature modeling, and to enhance the prior features of the center pixel in image patches, we propose an OC-Mamba block. As illustrated in Fig. 4, this block consists of an O-SSM and a CSSM.

1) One-Pixel Overlapped SSM: We integrate the center pixel prior and propose a local scanning mechanism, which can be denoted as O-SSM. Assume the input feature map is $\mathbf{F} \in \mathbb{R}^{P\times P\times C}$. We divide the feature map into four spatial subregions, each with a spatial size of $((P+1/2))\times((P+1/2))$. These subregions have a one-row overlapping area with their neighboring subregions. Each subregion is denoted as $\mathbf{E}_i \in \mathbb{R}^{((P+1/2))\times((P+1/2))\times C}$, where $i = 1, 2, 3, 4$.

For each subregion, a snake scanning mechanism is implemented: pixels are scanned row by row along the spatial dimension, and the scanning direction is reversed when transitioning to the next row. The four subregions commence scanning from the four corners of the feature map and conclude at the center pixel. Ultimately, the spatial dimensions are flattened, and the spectral channels of all spatial pixels are simultaneously loaded into the sequence.

After performing the scans in both directions, we flatten the spatial dimensions of all feature subregions and stack them together. To highlight the center pixel, suppress the edge pixels, and reduce the attention on overlapping areas, we introduce a decaying positional embedding for both scanning schemes. The specific formula is as follows:

$$b = \theta \cdot \log(a+1), \quad a > 0 \tag{15}$$

where θ represents a coefficient, and a is a positive integer denoting the sequence position. The positional embedding is directly added as a value to the corresponding positions in the feature map. Then these are input to the SSM block

$$\mathbf{F}_o = \mathrm{SSM}(\mathrm{OSS}(\mathbf{F}) \oplus \mathrm{DPE}(\mathrm{OSS}(\mathbf{F}))) \tag{16}$$

where $\mathbf{F}_o \in \mathbb{R}^{8\times((P+1/2))^2\times C}$ represents the output feature map after the S6 process, OSS stands for the overlapped snake scan, and DPE stands for the decay positional embedding.

Subsequently, we restore the feature subregions in each scanning direction to their original shape prior to scanning, as formulated below

$$\mathbf{F}_o^{i,j} = \mathrm{Split}\left(\mathrm{Reshape}\left(\mathbf{F}_o\right)\right), \quad i = 1, 2,\ j = 1, 2, 3, 4 \tag{17}$$

where i denotes the scanning direction, j represents the subregion index in each scanning direction, and $\mathbf{F}_o^{i,j} \in \mathbb{R}^{((P+1/2))\times((P+1/2))\times C}$. Subsequently, the subregions are concatenated along the spatial dimension to reconstruct the feature map $\mathbf{F}_{\mathrm{O-SSM}} \in \mathbb{R}^{(P+1)\times(P+1)\times C}$.

After that, a depthwise convolution with a kernel size of 2×2 is applied to downsample the feature map, restoring it to the original feature map dimensions to facilitate fusion with the feature map of CSSM. The operation is formulated as follows:

$$\mathbf{F}_O = \mathrm{Conv2d}_{2\times 2,\, g=C}\left(\mathbf{F}_{\mathrm{O-SSM}}\right) \tag{18}$$

where $\mathbf{F}_O \in \mathbb{R}^{P\times P\times C}$ represents the output feature map after SSM block, and $\mathrm{Conv2d}_{2\times 2,\, g=C}$ denotes a depthwise convolution with a kernel size of 2×2 and group size C.

2) Centering SSM: To extract the global spectral–spatial features of HSIs while fully leveraging the prior features of the center pixel, we propose the C-SSM. The input feature map $\mathbf{F} \in \mathbb{R}^{P\times P\times C}$ is first processed through a GAP operation for downsampling, resulting in $\mathbf{F}_c \in \mathbb{R}^{W\times W\times C}$, where W represents the spatial size after downsampling.

Subsequently, we integrate the center pixel prior into the feature map. Specifically, we obtain the CPV $\mathbf{v}_c \in \mathbb{R}^C$ from the output of the CS^{2}CA module and perform elementwise addition with the current feature map's CPV $\mathbf{v}_0 \in \mathbb{R}^C$ to fuse the spectral features from different branches.

Next, a 1-D convolution is applied to adjust the channel dimensions, facilitating the subsequent reshape operation. The

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CPV is then reshaped into a spatial size $\mathbf{V}_{\text{attn}} \in \mathbb{R}^{W\times W\times 1}$ and multiplied elementwise with the spatially downsampled feature map. The detailed operations are expressed as the following equation:

$$\begin{aligned}\mathbf{V}_{\text{attn}} &= \text{Reshape}\left(\text{Conv1d}\left(\mathbf{v}_c \oplus \mathbf{v}_0\right)\right)\\ \mathbf{F}'_c &= \mathbf{F}_c \otimes \mathbf{V}_{\text{attn}}.\end{aligned} \tag{19}$$

Next, we introduce a global snake scanning mechanism, starting from the top-left corner of the feature map and traversing the entire spatial dimension (consistent with the O-SSM). The scan terminates at the bottom-right corner. During this process, the spatial dimensions of the feature map are flattened into a single sequence, along with the spectral dimension, and then fed into the S6 module. The operation is formulated as follows:

$$\mathbf{F}_C = \text{SSM}\left(\text{GSS}\left(\mathbf{F}'_c\right)\right) \tag{20}$$

where $\mathbf{F}_C \in \mathbb{R}^{W^2\times C}$, and GSS represents the global snake scan.

Finally, a depthwise convolution with a kernel size of 3×3 is applied to upsample the feature map, restoring it to its original dimensions to facilitate fusion with the O-SSM feature map.

3) Fusion of O-SSM and C-SSM: To integrate local and global spectral–spatial features, we introduce a learnable weight λ that adaptively adjusts the balance between the two during the feature extraction process. The specific process is expressed as follows:

$$\mathbf{F}_{\text{OC-Mamba}} = \lambda \cdot \mathbf{F}_O \oplus (1-\lambda)\cdot \mathbf{F}_C \tag{21}$$

where λ represents the learnable weight.

F. Architecture of MSGC Block

For the feature maps from multiple branches, we perform two-stage spectral dimension integration in the MSGC module. As illustrated in the MSGC on the right side of Fig. 1, in each stage we employ two parallel grouped convolutions with a spatial size of 1×1. The first stage involves coarse grouping with group sizes of 2 and 4, while the second stage involves fine grouping with group sizes of 8 and 16. Assume the input feature map is $\mathbf{F}_0 \in \mathbb{R}^{P\times P\times C}$. The specific process is listed below

$$\begin{aligned}\mathbf{F}_i &= \text{ReLU}\left(\text{BN}\left(\text{Conv2d}_{1\times 1, g=2*(4^{i-1})}\left(\mathbf{F}_{i-1}\right)\right)\right)\\ &\quad \oplus \text{ReLU}\left(\text{BN}\left(\text{Conv2d}_{1\times 1, g=4*(4^{i-1})}\left(\mathbf{F}_{i-1}\right)\right)\right)\\ \mathbf{F}_i^{\text{MSGC}} &= \text{LeakyReLU}\left(\text{BN}\left(\mathbf{F}_{i-1}\right)\right)\end{aligned} \tag{22}$$

where $i = 1, 2$ represents the stage number, and $\mathbf{F}_i^{\text{MSGC}} \in \mathbb{R}^{P\times P\times C_i}$ stands for the output feature map of ith stage.

IV. Experimental Results and Discussions

This section is organized as follows. Section IV-A describes four hyperspectral datasets used to assess the SliMamba model, including Houston, Trento, WHU-Hi-HongHu, and Fanglu Tea Farm. Section IV-B analyzes the impact of different hyperparameters (patch size, PCA-reduced channel numbers, and SSC output channel numbers) on the model's performance. Section IV-C compares various convolution methods and presents an ablation study on the SSC. Section IV-D describes an ablation study of the CS^{2}CA mechanism. Section IV-E reports an ablation study of the OC-Mamba module and compares it with the multihead self-attention mechanism. Finally, Section IV-G compares the classification performance and computational complexity of our model with several state-of-the-art network architectures.

To begin with, we present the parameter configurations used in our experiments. The size of the input feature map is initialized as $C \times 9 \times 9$, where C represents the number of spectral bands in the dataset. In the dimensionality-reduction branch, following the PCA operation, the number of channels is reduced to 32. The data values are normalized to the range $[0, 1]$. Furthermore, data augmentation techniques are applied to the input patches. Specifically, each patch is first horizontally or vertically flipped, and then randomly rotated by 90°, 180°, or 270°. During training, we use the SGD optimizer to update the parameters over 200 epochs. The learning rate is set to 0.1, with a weight decay coefficient of 0.0001 and a momentum of 0.9. To minimize the impact of randomness and ensure the reliability of the results, each experiment is conducted ten times and the average result is recorded. The overall accuracy (OA), average accuracy (AA), and κ coefficient are employed to assess the performance of different classification methods. OA denotes the proportion of correctly predicted labels out of the total number of labels, AA represents the average accuracy across all classes, and the κ coefficient measures the agreement between predicted and actual labels. The experimental environment is configured as follows: Python 3.7, PyTorch 2.2.0, CUDA 11.7, and an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory.

A. Hyperspectral Datasets

In our experiments, four benchmark hyperspectral datasets are used to evaluate the performance of our proposed SliMamba model. The specific classes of the dataset, the corresponding colors in the subsequent classification maps, and the division of the training, validation, and test sets are detailed in Table I.

1) Houston: The Houston dataset, provided by the 2013 GRSS Data Fusion Contest, was collected using the ITRES CASI-1500 sensor over the University of Houston campus and its surrounding areas. The spectrometer captures wavelengths ranging from 0.38 to 1.05 μm. The dataset consists of 144 spectral bands and 340×1905 pixels, with a spatial resolution of 2.5 m/pixel. A total of 16,373 sample pixels are categorized into 15 challenging classes.

2) Trento: The Trento dataset was collected using the AISA Eagle sensor, which captures 63 spectral bands and was acquired over rural areas in southern Trento, Italy. The dataset has a spatial size of 166×600 pixels, with a spatial resolution of 1 m/pixel. It contains six distinct classes.

3) WHU-Hi-HongHu: The WHU-Hi-HongHu dataset [62] was acquired using a 17-mm focal length Headwall Nano-Hyperspec imaging sensor mounted on a DJI Matrice 600 Pro UAV platform. The dataset contains 22 challenging ground object categories, with an image size of 940×475 pixels and 270 spectral bands covering wavelengths from 400 to 1000 nm. The spatial resolution of the UAV-borne HSI is approximately 0.043 m.

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TABLE I

CLASS NAMES, CLASS COLORS FOR CLASSIFICATION MAP, AND SAMPLE NUMBERS OF TRAINING SET, VALIDATION SET, AND TEST SET FOR EACH CLASS IN THE HOUSTON, TRENTO, WHU-HI-HONGHU, AND FANGLU TEA FARM DATASETS

	Houston				Trento				WHU-Hi-HongHu				Fanglu Tea Farm			
No.	Class Name	Tr.	Val.	Te.	Class Name	Tr.	Val.	Te.	Class Name	Tr.	Val.	Te.	Class Name	Tr.	Val.	Te.
1	Healthy grass	125	13	1114	Apples	81	8	3945	Red roof	281	28	13732	Masson pine	116	29	5661
2	Stressed grass	126	13	1116	Buildings	58	6	2839	Road	70	7	3435	Bamboo forest	46	12	2260
3	Synthetic grass	69	7	620	Ground	9	1	469	Bare soil	436	44	21341	Tea tree	569	142	27717
4	Trees	125	12	1107	Woods	182	18	8923	Cotton	3265	327	159693	Reed	4	1	209
5	Soil	124	12	1105	Vineyard	210	21	10270	Cotton firewood	125	12	6081	Rice	136	34	6639
6	Water	32	3	289	Roads	63	7	3104	Rape	891	89	43577	Sweet potato	16	4	797
7	Residential	126	13	1129					Chinese cabbage	482	48	23573	Coriander	9	2	418
8	Commercial	125	13	1107					Pakchoi	81	8	3965	Waste grass	38	9	1814
9	Road	125	13	1114					Cabbage	216	22	10581	Ponds	122	31	5988
10	Highway	123	12	1092					Tuber mustard	248	25	12121	Building/road	18	5	888
11	Railway	124	12	1099					Brassica parachinensis	220	22	10773				
12	Parking Lot1	123	12	1097					Brassica chinensis	179	18	8757				
13	Parking Lot2	46	5	418					Small Brassica chinensis	450	45	22012				
14	Tennis Court	42	4	381					Lactuca sativa	147	15	7194				
15	Running Track	66	7	588					Celtuce	20	2	980				
16									Film covered lettuce	145	15	7102				
17									Romaine lettuce	60	6	2944				
18									Carrot	65	6	3146				
19									White radish	175	17	8520				
20									Garlic sprout	70	7	3409				
21									Broad bean	26	3	1299				
22									Tree	81	8	3951				
Total		1501	151	13376		603	61	29550		7733	774	378186		1074	269	52391

4) Fanglu Tea Farm: The Fanglu Tea Farm dataset was collected from the Fanglu Tea Plantation in Changzhou, China, using the aerosol pushbroom hyperspectral imager (PHI). The dataset has a spatial resolution of 2.25 m, consists of 348 × 512 pixels, and includes 80 spectral bands with wavelengths ranging from 417 to 855 nm. It contains 53 734 labeled pixels, covering ten classes of interest.

B. Hyperparameters Analysis

In this section, we analyze the impact of three hyperparameters on the model's performance: patch size, the number of PCA output channels, and the number of SSC output channels.

1) Patch Size: We systematically evaluated the effect of input patch size on model performance using seven metrics—OA, AA, κ, parameter count, FLOPs, and inference time—across four datasets, with patch sizes set to 5, 9, 13, 17, 21, and 25 [see Fig. 5(a)–(g)]. As shown in Fig. 5(a)–(d), OA, AA, and κ on the Houston, Trento, and Fanglu Tea Farm datasets generally increase then decrease with patch size, peaking at 9. On the WHU-Hi-HongHu dataset, these metrics show a slow, steady improvement as patch size increases. However, overly large patch sizes, while expanding the receptive field, do not enhance performance and may cause spatial information interference, resulting in decreased classification accuracy.

Fig. 5(e)–(g) further reveals that model size (in terms of parameter count and FLOPs) and inference time both grow substantially with increasing patch size, due to the feature map channel size in the SSC module scaling quadratically.

Qualitative analysis on the Trento dataset (Fig. 6) demonstrates that larger patch sizes (e.g., 21) can cause mixing of distinct land cover types within receptive fields, resulting in significant misclassification. In contrast, a patch size of 9 achieves finer discrimination between classes by limiting spatial ambiguity.

TABLE II

SENSITIVITY ANALYSIS OF THE PROPOSED METHOD WITH VARYING PCA-REDUCED CHANNEL NUMBERS ON THE HOUSTON AND WHU-HI-HONGHU DATASETS

	Houston			WHU-Hi-HongHu		
PCA	OA	AA	κ	OA	AA	κ
8	98.26±0.12	98.03±0.06	98.21±0.12	95.9±0.11	91.26±0.36	95.22±0.13
16	98.56±0.21	98.21±0.26	98.51±0.22	96.48±0.17	92.36±0.42	95.98±0.16
32	**98.94±0.23**	**98.75±0.25**	**98.85±0.25**	**96.96±0.19**	92.99±0.46	**96.16±0.24**
64	98.88±0.20	98.60±0.12	98.71±0.18	96.94±0.12	**93.03±0.43**	96.13±0.17
128	98.71±0.17	98.51±0.08	98.68±0.10	96.86±0.21	92.76±0.51	96.12±0.12

Based on these results, we select a patch size of 9 for our model to strike a balance between receptive field and classification accuracy.

2) PCA-Reduced Channel Numbers: In the dimensionality-reduction branch, PCA is employed to reduce the spectral dimensionality of the HSI data. To determine the optimal number of principal components, we evaluated model performance on the Houston and WHU-Hi-HongHu datasets with $L = \{8, 16, 32, 64, 128\}$, as shown in Table II. The results show that increasing L initially enhances OA, AA, and κ, with peak performance typically observed at $L = 32$ or 64. For instance, on the Houston dataset, the model achieves its highest OA (98.94% ± 0.23), AA (98.75% ± 0.25), and κ (98.85% ± 0.25) when $L = 32$, with similar trends on WHU-Hi-HongHu. However, further increases in L beyond this point result in performance degradation, likely due to the Hughes phenomenon, where excessive features lead to overfitting and information redundancy. Therefore, we select 32 principal components to achieve an effective trade-off between classification accuracy and computational efficiency.

3) SSC Output Channel Numbers: The number of output channels in the SSC module effectively controls the model's spectral representation capacity and can be regarded as a form

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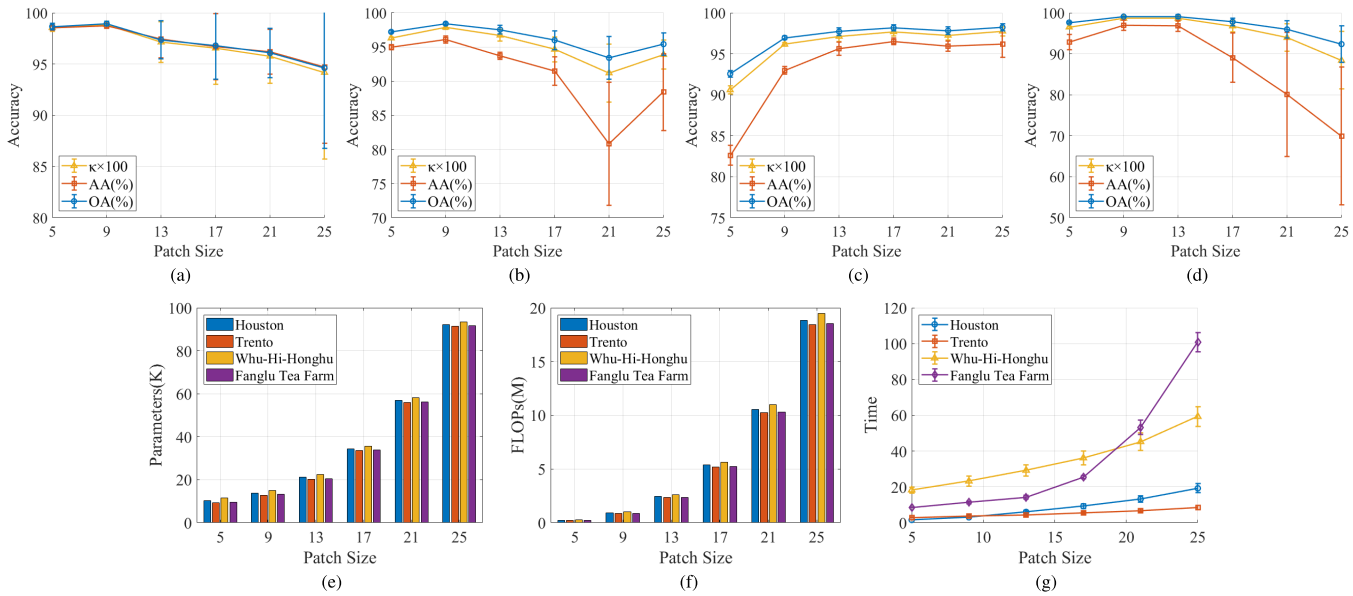


Fig. 5. (a)–(d) Model's performance in terms of OA, AA, and κ for different patch sizes across four datasets: (a) Houston, (b) Trento, (c) WHU-Hi-HongHu, and (d) Fanglu Tea Farm. (e)–(g) Model's parameters, FLOPs, and inference time for varying patch sizes across the four datasets: (e) parameters, (f) FLOPs, and (g) inference time.

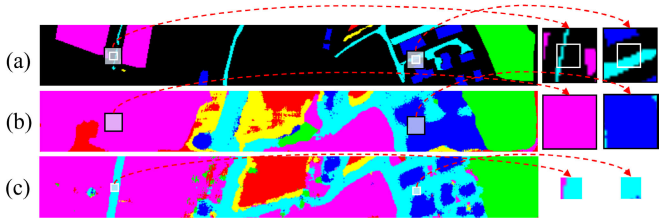


Fig. 6. Visualization of classification maps on the Trento dataset under different patch sizes. The white box indicates a patch size of 9, and the black box indicates a patch size of 21. (a) Ground truth. (b) Classification map (OA = 93.41%) with patch size 21. (c) Classification map (OA = 98.42%) with patch size 9.

TABLE III

SENSITIVITY ANALYSIS OF THE PROPOSED METHOD WITH VARYING SSC OUTPUT CHANNEL NUMBERS ON THE HOUSTON AND WHU-HI-HONGHU DATASETS

SSC	Houston			WHU-Hi-HongHu		
	OA	AA	κ	OA	AA	κ
8	98.03±0.09	97.93±0.05	98.01±0.06	95.43±0.12	90.26±0.46	95.12±0.23
16	98.46±0.21	98.13±0.12	98.42±0.18	96.68±0.15	91.43±0.58	95.78±0.21
32	98.94±0.23	**98.75±0.25**	98.85±0.25	**96.96±0.19**	**92.99±0.46**	**96.16±0.24**
64	**98.98±0.25**	98.73±0.28	**98.91±0.28**	96.74±0.12	92.73±0.43	95.73±0.38
128	98.92±0.19	98.72±0.18	98.87±0.12	96.65±0.21	92.26±0.54	96.22±0.16

of dimensionality reduction. To determine the optimal setting, we evaluated model performance on the Houston and WHU-Hi-HongHu datasets by varying the number of SSC output channels $C = \{8, 16, 32, 64, 128\}$, as shown in Table III. The results show that increasing C generally leads to improvements in OA, AA, and κ, with peak performance observed at $C = 32$ or 64. For example, on Houston, the highest OA (98.98% $\pm$ 0.25), AA (98.75% $\pm$ 0.28), and κ (98.91% $\pm$ 0.26) are achieved at $C = 64$. On WHU-Hi-HongHu, optimal results are observed at $C = 32$ (OA: 96.96% $\pm$ 0.19, AA: 92.99% $\pm$ 0.46, and κ: 96.16% $\pm$ 0.24). Further increasing the number of output channels does not yield additional gains and may even cause a slight decline in accuracy, potentially due to increased feature redundancy and overfitting. Therefore, selecting 32 channels in the SSC module provides an effective trade-off between expressive power and model generalization.

C. Analysis of the SSC Block

In this section, we provide a comprehensive evaluation of the proposed SSC module. SSC is benchmarked against standard convolution (SC) and depthwise convolution (DSC) with a 3×3 kernel, and we further investigate the influence of different dilation rates in Step ③. We also examine the effect of introducing the CPV from an auxiliary branch by comparing variants with and without this component.

As shown in Table IV, the results are summarized in terms of OA, parameter count, and FLOPs across four standard datasets.

Replacing SSC with SC leads to a significant drop in OA for all datasets. For example, the OA on WHU-Hi-HongHu decreases to 83.83%. This replacement also sharply increases the number of parameters and FLOPs due to the absence of channel grouping. When using DSC, the OA is generally lower than that of SSC on all datasets. Notably, SSC not only achieves higher accuracy but also further reduces parameters and computational cost, owing to its ability to effectively capture cross-channel spectral relationships.

Regarding dilation, we find that a rate of 3 is optimal for the Houston dataset, while a rate of 2 works best for the others, reflecting different spectral properties across datasets.

The CPV branch, which provides the center pixel vector as complementary information, also proves essential: Removing CPV consistently lowers OA, highlighting its value for robust classification.

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TABLE IV

COMPARISON WITH TRADITIONAL CONVOLUTION METHODS AND ANALYSIS ON CONVOLUTION TYPES IN SSC ACROSS FOUR DATASETS

		Houston			Trento			WHU-Hi-HongHu			Fanglu Tea Farm		
		OA(%)	Params(K)	FLOPs(M)	OA(%)	Params(K)	FLOPs(M)	OA(%)	Params(K)	FLOPs(M)	OA(%)	Params(K)	FLOPs(M)
SC	-	98.38±0.29	46.76	3.7	98.33±0.36	45.5	3.65	83.83±3.24	48.05	3.79	98.63±0.05	45.76	3.61
DSC	-	98.59±0.44	14.76	1.12	98.14±0.28	13.5	1.02	96.54±0.29	16.03	1.2	98.77±0.25	13.76	1.03
SSC (w CPV)	D=1	99.07±0.20	13.84	0.93	98.27±0.27	12.9	0.88	96.51±0.29	15.42	1.06	98.80±0.08	13.16	0.89
	D=2	98.94±0.23	13.84	0.93	**98.42±0.21**	12.9	0.88	**96.96±0.19**	15.42	1.06	**99.09±0.16**	13.16	0.89
	D=3	**99.24±0.21**	13.84	0.93	98.12±0.47	12.9	0.88	96.62±0.18	15.42	1.06	98.82±0.21	13.16	0.89
SSC (w/o CPV)	D=2	98.88±0.18	13.84	0.93	98.37±0.23	12.9	0.88	96.61±0.23	15.42	1.06	98.80±0.04	13.16	0.89

TABLE V

ABLATION ON CS^{2}CA BLOCK

			Houston			Trento			WHU-Hi-HongHu			Fanglu Tea Farm		
CA	SA	CPV	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
		✓	97.98±0.24	97.90±0.30	97.90±0.26	97.60±0.44	95.50±0.94	96.33±0.59	96.54±0.16	91.63±0.61	95.65±0.20	98.82±0.19	96.27±1.11	98.25±0.28
	✓		98.06±0.40	97.89±0.48	97.99±0.43	97.67±0.35	95.59±0.74	96.19±0.47	96.35±0.09	92.36±1.03	95.39±0.12	98.19±0.30	96.42±0.80	97.58±0.40
✓		✓	98.41±0.19	98.34±0.31	98.42±0.21	97.81±0.46	95.71±1.41	97.08±0.62	96.58±0.19	92.45±0.24	95.68±0.23	98.94±0.17	96.71±0.69	98.43±0.25
✓	✓		98.38±0.39	98.23±0.62	98.38±0.42	97.76±0.49	95.68±1.02	97.15±0.66	96.58±0.20	92.63±0.87	95.68±0.26	98.88±0.18	96.08±1.92	98.34±0.27
✓	✓	✓	**98.94±0.23**	**98.75±0.25**	**98.85±0.25**	**98.42±0.21**	**96.08±0.54**	**97.89±0.28**	**96.96±0.19**	**92.99±0.46**	**96.16±0.24**	**99.09±0.16**	**96.95±1.27**	**98.65±0.24**

TABLE VI

ABLATION STUDY ON OC-MAMBA AND COMPARISON WITH MHSA AND PLAIN MAMBA

		Houston			Trento			WHU-Hi-HongHu			Fanglu Tea Farm		
		OA(%)	Params(K)	FLOPs(M)	OA(%)	Params(K)	FLOPs(M)	OA(%)	Params(K)	FLOPs(M)	OA(%)	Params(K)	FLOPs(M)
O-SSM (w/o C-SSM)	OSS #1(w/o DPE)	98.16±0.15	12.48	0.85	97.67±0.25	11.52	0.8	96.15±0.18	14.06	0.98	98.10±0.20	11.8	0.81
	OSS #1(w DPE)	98.20±0.12	12.48	0.85	97.71±0.21	11.52	0.8	96.33±0.20	14.06	0.98	98.18±0.25	11.8	0.81
	OSS #2(w/o DPE)	98.12±0.15	12.48	0.85	97.50±0.18	11.52	0.8	96.14±0.16	14.06	0.98	98.15±0.22	11.8	0.81
	OSS #2(w DPE)	98.30±0.12	12.48	0.85	97.49±0.17	11.52	0.8	96.36±0.19	14.06	0.98	98.14±0.28	11.8	0.81
	OSS ALL(w/o DPE)	98.50±0.12	12.96	0.88	97.82±0.21	12	0.83	96.26±0.24	14.54	1.01	98.24±0.48	12.28	0.84
	OSS ALL(w DPE)	98.54±0.29	12.96	0.88	98.02±0.34	12	0.83	96.50±0.44	14.54	1.01	98.29±0.45	12.28	0.84
C-SSM (w/o O-SSM)	GSS(C=3)	98.23±0.13	12.7	0.85	97.33±0.62	11.74	0.8	96.25±0.13	14.28	0.98	98.28±0.10	12.02	0.81
	GSS(C=5)	98.39±0.17	12.88	0.87	97.62±0.54	11.92	0.82	96.55±0.29	14.46	1	98.45±0.31	12.2	0.83
	GSS(C=7)	98.46±0.20	13.08	0.89	97.46±0.32	12.12	0.84	96.32±0.12	14.67	1.02	98.35±0.11	12.41	0.85
	GSS(C=9)	98.32±0.21	13.3	0.91	97.65±0.12	12.34	0.87	96.52±0.22	14.89	1.05	98.44±0.25	12.63	0.88
O-SSM (w C-SSM GSS(C=5))	OSS #1(w/o DPE)	98.63±0.19	13.36	0.9	97.78±0.23	12.4	0.85	96.27±0.14	14.94	1.03	98.65±0.22	12.68	0.86
	OSS #1(w DPE)	98.65±0.22	13.36	0.9	97.79±0.16	12.4	0.85	96.42±0.23	14.94	1.03	98.64±0.22	12.68	0.86
	OSS #2(w/o DPE)	98.59±0.21	13.36	0.9	97.85±0.22	12.4	0.85	96.31±0.17	14.94	1.03	98.60±0.20	12.68	0.86
	OSS #2(w DPE)	98.55±0.22	13.36	0.9	98.04±0.23	12.4	0.85	96.38±0.33	14.94	1.03	98.61±0.22	12.68	0.86
	OSS ALL(w/o DPE)	98.93±0.20	13.84	0.93	98.16±0.48	12.9	0.88	96.54±0.11	15.42	1.06	98.68±0.60	13.16	0.89
C-SSM (w O-SSM (OSS All (w DPE)))	GSS(C=3)	98.75±0.19	13.66	0.91	97.75±0.13	12.7	0.86	96.78±0.12	15.24	1.04	98.77±0.21	12.98	0.87
	GSS(C=5)	**98.94±0.23**	13.84	0.93	98.42±0.21	12.9	0.88	**96.96±0.19**	15.42	1.06	**99.09±0.16**	13.16	0.89
	GSS(C=7)	98.87±0.21	14.04	0.95	**98.52±0.21**	13.08	0.90	96.88±0.22	15.63	1.08	98.82±0.20	13.37	0.91
	GSS(C=9)	98.91±0.21	14.26	0.98	98.15±0.22	13.3	0.93	96.76±0.23	15.85	1.11	98.71±0.22	13.59	0.94
MHSA	plain MHSA	97.28±2.59	31.93	2.46	97.05±0.37	33.99	2.4	96.36±0.33	33.18	2.54	98.40±0.67	31.25	2.4
Mamba	plain 2D Scanning	94.46±5.40	19.67	1.46	97.01±3.31	18.74	1.41	96.68±0.14	20.9	1.54	96.65±1.97	19	1.42

Overall, these results validate the effectiveness of SSC, the importance of proper dilation configuration, and the supportive role of the CPV branch in HSIC.

D. Ablation on CS^{2}CA Block

The CS^{2}CA block facilitates cross-attention (CA) between the GSV and the CPV within the SSC module. Its output includes spatial attention (SA), which is applied to the SSC output feature map, and the CPV, which is fed to the OC-Mamba module. To quantify the contributions of CA, SA, and CPV, we conducted extensive ablation studies on four datasets using OA, AA, and κ as evaluation metrics (see Table V). The first two rows illustrate the effect of CA. Without CA, using only SA or CPV leads to suboptimal performance, likely because there is no interaction between spatial features (GSV) and spectral features (CPV). Incorporating CA substantially improves all metrics. For example, on Houston, OA increases by 0.43% (from 97.98% to 98.41%) when using both CA and CPV compared with using CPV alone. On WHU-Hi-HongHu, although the OA improvement is small (0.04%), AA increases by 0.82%. With CA included, adding either SA or CPV yields further improvements, as shown in the last three rows. Specifically, SA increases OA by about 0.4 to 0.5% on the first three datasets and raises AA by 0.54% (from 92.45% to 92.99%) on WHU-Hi-HongHu. It also increases the κ coefficient on Trento by 0.81 (from 97.08 to 97.89). Similarly, incorporating the CPV module brings OA gains of approximately 0.4% on most datasets. For instance, on Fanglu Tea Farm, AA improves from 96.08% to 96.95%, corresponding to a gain of 0.87%. In summary, these results underscore the essential role of CA and demonstrate that both SA and center pixel information provide complementary benefits to model performance.

E. Ablation on OC-Mamba Block

Table VI systematically presents the ablation studies of the OC-Mamba module, quantitatively analyzing the functional contributions of each core component. The study specifically

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examines the two serpentine scanning strategies in the Overlapped SSM (O-SSM) branch (OSS #1 and OSS #2), the decaying positional encoding (DPE), and the downsampling size C in the C-SSM branch.

For the O-SSM branch, the results show that OSS #1 and OSS #2 achieve highly similar classification accuracies across datasets. For example, on the Houston dataset, the OA for OSS #1 and OSS #2 (without DPE) reach 98.16% and 98.12%, respectively, indicating that both scanning strategies can effectively model spatial dependencies. The incorporation of DPE consistently leads to further performance gains, especially on challenging datasets such as WHU-Hi-HongHu, where the OA increases from 96.15% to 96.33%. These findings suggest that the two scanning strategies capture complementary neighborhood relationships from different directions, and their fusion allows richer spatial context representation. Meanwhile, DPE suppresses superfluous responses to distant positions, alleviating the overconcentration of attention in the central regions caused by repeated scanning, thus preserving the discriminative power for edge pixels. The combination of both scanning strategies with DPE yields the best performance, with Houston reaching an OA of 98.54%.

For the C-SSM branch, the impact of the downsampling size C on global feature aggregation is evident. As C increases from 3 to 5 or 7, OA generally rises and then stabilizes. For instance, in the Houston dataset, OA climbs from 98.23% at $C = 3$ to 98.46% at $C = 7$, while Fanglu Tea Farm achieves a peak OA of 98.45% at $C = 5$. These observations indicate that an appropriately chosen downsampling window efficiently suppresses redundant information during feature aggregation and strengthens global representation in central regions.

When O-SSM and C-SSM branches are jointly modeled, notable and consistent performance improvements are observed across all benchmark datasets. Further incorporation of DPE offers sustained enhancement, with OA on the Trento dataset improving from 97.85% to 98.04% and WHU-Hi-HongHu from 96.31% to 96.38%. The complete OC-Mamba configuration (O-SSM[ALL] + C-SSM + DPE, $C = 5$) delivers the highest OA in most cases, such as 98.94% on Houston, 96.96% on WHU-Hi-HongHu, and 99.09% on Fanglu Tea Farm, while Trento achieves its peak (98.52%) at $C = 7$. For balanced stability and generalization in further experiments, $C = 5$ is selected as the default setting.

In terms of model complexity, both O-SSM and C-SSM are lightweight designs with minimal parameters and FLOPs overhead. Notably, DPE in the O-SSM branch is implemented without introducing any additional parameters, greatly improving the deployability of the model.

Compared with baseline models (the last two rows of the table), OC-Mamba achieves significantly higher OA with fewer parameters and lower FLOPs on all datasets. Although plain Mamba benefits from linear computational complexity, its global 2-D scanning leads to excessive uniformity and insufficient spatial discrimination, resulting in reduced accuracy (e.g., OA of only 94.46% on Houston). MHSA imposes substantial computational and parameter burdens (both more than twice that of OC-Mamba) due to high-dimensional QKV projections and exhaustive pairwise spatial interactions, but still fails to deliver competitive accuracy. This further confirms the effectiveness of OC-Mamba: the dual-branch design excels at multiscale spatial relationship modeling, balancing fine-grained discrimination with computational efficiency.

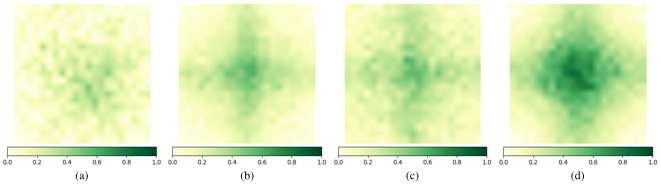


Fig. 7. Visualization of ERF within OC-Mamba block. (a) C-SSM. (b) O-SSM #1. (c) O-SSM #2. (d) OC-Mamba.

TABLE VII
ABLATION ON DIFFERENT MODULE OF THE NETWORK IN FANGLU TEA FARM DATASET

	SA-SSM	SSC	CS^{2}CA	OC-Mamba	OA(%)	Params(K)	FLOPs(M)	Time(s)
1	✓				95.22	6.87	0.5	8.36
2		✓	✓		96.68	10.84	0.76	9.33
3				✓	96.97	8.23	0.59	8.66
4	✓	✓	✓		97.14	11.32	0.78	9.68
5	✓	✓		✓	98.75	13.06	0.86	11.22
6		✓	✓	✓	98.60	12.68	0.87	11.09
7	✓			✓	97.24	8.71	0.61	9.16
8	✓	✓	✓	✓	**99.09**	13.16	0.89	11.34

To further highlight the unique advantages of the proposed OC-Mamba module, we conduct a visual ablation analysis based on the effective receptive field (ERF), in addition to quantitative metrics. As illustrated in Fig. 7, the ERF visualizations of each subcomponent reveal their complementary behaviors in capturing spatial information. Specifically, C-SSM concentrates on the central region of the feature map, whereas the two O-SSM scanning schemes (OSS #1 and OSS #2) exhibit cross-shaped receptive fields that diffuse outward from the center along different directions. After fusion, the receptive field of the OC-Mamba module extends multidirectionally from the central pixel and eventually covers the entire spatial domain. This comprehensive and multidirectional receptive field not only underscores the critical role of the central pixel as the label location, but also enables the model to effectively perceive and model similarities among neighboring objects, leading to more precise boundary delineation. At the same time, the global coverage provided by the combined receptive field substantially enhances the module's capacity for context aggregation and holistic scene understanding, thereby fully demonstrating the architectural strengths of OC-Mamba.

F. Ablation on Different Module of the Network

To thoroughly analyze the contributions of each module in the proposed network, as shown in Table VII, we conducted comprehensive ablation experiments on the SA-SSM, SSC, CS^{2}CA, and OC-Mamba blocks using the Fanglu Tea Farm dataset. The model was evaluated using OA, parameters, FLOPs, and inference time as metrics.

First, by comparing rows 6 and 8 in the table for SA-SSM, we observe that employing SA-SSM improves the model's OA by 0.49%, while resulting in a 3.7% increase in parameters, a 2.3% rise in FLOPs, and an additional 0.26 s in inference time. Second, due to the certain coupling nature of SSC and CS^{2}CA, a comparison between rows 7 and 8 shows that utilizing these

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TABLE VIII

OAs[%], AAs[%], $\kappa \times 100$, TRAINING TIME, AND TESTING TIME STATISTIC FOR THE HOUSTON SCENE

	CNN-based			Transformer-Based			SSM-Based							
Class	CVSSN TCSVT2022	MSCA-DC GRSL2023	HKNAS3D TNNLS2023	SSFTT TGRS2022	MorphFormer TGRS2023	CMT TGRS2024	MambaHSI TGRS2024	MiM IJON2024	3DSS-Mamba TGRS2024	IGSS-Mamba TGRS2024	HyperMamba TGRS2024	MambaLG TGRS2024	SpiralMamba TGRS2025	SliMamba Proposed
1	96.97±2.27	99.19±1.57	88.11±3.16	96.16±3.66	97.65±1.29	98.64±1.11	98.74±2.20	97.77±0.22	**99.58±0.61**	97.15±0.64	98.85±0.79	98.69±0.41	99.12±0.44	98.83±1.21
2	98.80±1.51	96.74±2.16	90.18±5.27	98.33±2.93	97.19±4.00	99.28±0.38	98.26±1.10	99.28±0.21	94.44±1.46	98.96±0.43	99.03±0.64	98.71±0.84	99.09±0.39	**99.57±0.16**
3	99.97±0.06	99.94±0.13	98.61±0.43	**100.00±0.00**	99.94±0.13	**100.00±0.00**	99.97±0.06	99.61±0.16	99.20±0.72	99.65±0.19	99.84±0.25	99.97±0.06	**100.00±0.00**	99.94±0.08
4	97.11±2.43	94.09±0.26	83.67±3.20	98.57±2.19	**99.19±0.86**	98.77±1.36	98.57±1.07	98.63±0.48	94.53±3.98	97.11±1.54	96.30±2.74	96.98±1.05	96.21±1.84	98.79±1.20
5	99.35±0.22	99.66±0.64	99.04±1.56	99.20±1.25	99.62±0.47	**100.00±0.00**	99.98±0.04	**100.00±0.00**	96.01±0.36	99.98±0.04	**100.00±0.00**	99.98±0.04	99.78±0.39	**100.00±0.00**
6	97.74±2.50	87.75±3.77	81.87±5.05	98.41±2.69	99.38±0.83	**99.58±0.83**	94.34±5.76	92.73±4.24	92.12±1.83	98.69±2.63	98.75±1.02	97.79±2.84	97.79±2.69	97.30±2.11
7	95.78±0.91	94.21±2.90	91.85±4.36	95.93±2.41	97.70±1.30	94.93±2.06	98.46±1.04	**98.72±1.21**	94.40±3.66	95.25±1.59	96.76±0.45	97.11±2.02	97.95±1.15	98.62±0.62
8	**98.94±0.45**	88.55±4.94	93.39±1.46	95.10±4.72	97.67±1.27	94.76±4.68	92.82±3.37	96.30±1.59	94.38±2.78	98.08±1.42	96.87±1.69	97.09±2.30	98.36±1.17	98.19±1.25
9	**97.90±0.92**	93.43±3.40	90.97±1.68	96.57±2.07	96.14±1.43	95.94±2.58	95.98±4.34	95.87±2.23	92.88±2.94	90.81±8.89	97.88±0.80	96.79±2.00	97.16±1.53	97.63±1.19
10	97.79±0.83	97.84±3.87	97.53±3.97	98.35±1.78	98.55±1.11	94.91±6.38	98.44±1.65	99.34±1.05	93.91±2.13	99.51±0.51	98.04±1.72	99.08±0.95	98.96±0.52	**99.56±0.20**
11	98.39±1.06	95.65±2.50	95.65±4.13	93.98±4.64	96.62±1.12	92.78±6.80	95.51±3.19	**99.96±0.07**	95.69±1.54	98.76±2.03	98.69±1.31	98.67±2.21	98.40±1.53	99.91±0.12
12	96.51±2.65	94.49±6.18	98.41±1.07	93.40±4.86	98.18±0.31	97.12±2.91	98.01±0.69	96.86±2.32	**99.84±0.21**	99.02±0.78	95.88±4.80	97.16±3.16	98.80±0.63	99.22±0.37
13	96.98±1.05	91.63±2.28	91.39±3.09	95.89±3.14	95.89±1.24	96.70±2.62	94.08±1.31	89.00±2.35	**99.05±1.40**	91.05±3.36	93.88±2.57	93.78±2.05	94.16±0.94	93.88±3.20
14	99.69±0.50	**100.00±0.00**	**100.00±0.00**	99.79±0.42	99.84±0.21	99.95±0.10	**100.00±0.00**	99.79±0.20	95.90±0.66	99.84±0.31	99.95±0.10	99.42±0.56	99.95±0.10	**100.00±0.00**
15	98.61±1.34	**100.00±0.00**	99.29±1.10	99.93±0.08	**100.00±0.00**	99.97±0.07	96.33±0.54	99.90±0.20	95.16±0.75	99.97±0.07	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.86±0.17
OA(%)	97.89±0.56	95.64±0.99	93.33±1.07	96.98±1.37	98.07±0.49	97.56±0.79	97.44±1.26	98.04±0.26	95.52±0.69	97.56±0.68	97.98±0.41	98.10±0.79	98.43±0.27	**98.94±0.23**
AA(%)	98.04±0.52	95.54±0.92	93.33±1.14	97.31±1.13	98.24±0.36	98.02±0.31	97.30±1.29	97.58±0.38	96.39±4.24	97.59±0.61	98.05±0.37	98.08±0.61	98.38±0.29	**98.75±0.25**
$\kappa \times 100$	97.72±0.6	95.29±1.07	92.78±1.16	96.73±1.48	97.91±0.53	96.94±0.99	98.03±1.23	97.88±0.28	96.02±2.22	97.36±0.73	97.81±0.45	97.95±0.86	98.30±0.29	**98.85±0.25**
Training Time(s)	**13.74±0.49**	27.30±1.14	44.32±2.32	17.83±0.23	56.32±3.3	23.33±0.84	66.48±6.83	99.23±5.23	21.83±0.77	68.59±2.59	43.26±2.26	45.69±2.63	95.26±5.03	24.30±1.14
Test Time(s)	**2.31±0.054**	4.03±1.12	6.32±1.3	3.11±0.21	4.34±1.09	3.73±0.25	4.92±0.99	6.27±2.11	4.89±0.33	6.14±1.98	5.26±1.03	4.95±0.26	6.18±1.06	3.03±1.12
Parameters(K)	253.36	46.59	103.04	673.74	154.74	103.23	412.23	106.73	**10.51**	57.51	159.19	189.63	1950.26	13.84
FLOPs(M)	654.89	3.71	87.56	54.54	17.58	5.2	486.76	130.22	7.12	9.52	9.72	20.56	56.67	**0.93**

TABLE IX

OAs[%], AAs[%], $\kappa \times 100$, TRAINING TIME, AND TESTING TIME STATISTIC FOR THE TRENTO SCENE

	CNN-based			Transformer-Based			SSM-Based							
Class	CVSSN TCSVT2022	MSCA-DC GRSL2023	HKNAS3D TNNLS2023	SSFTT TGRS2022	MorphFormer TGRS2023	CMT TGRS2024	MambaHSI TGRS2024	MiM IJON2024	3DSS-Mamba TGRS2024	IGSS-Mamba TGRS2024	HyperMamba TGRS2024	MambaLG TGRS2024	SpiralMamba TGRS2025	SliMamba Proposed
1	**99.82±0.17**	98.90±1.06	99.26±0.30	99.19±0.75	99.39±0.41	99.17±0.44	99.77±0.15	98.62±0.39	98.63±0.92	98.53±0.50	99.09±1.15	99.46±0.31	99.57±0.32	99.00±0.30
2	91.59±3.52	90.38±5.21	85.45±7.20	80.00±4.24	83.18±3.83	91.72±3.33	91.24±1.39	91.44±3.85	85.27±9.71	83.56±9.07	83.28±5.97	78.34±16.43	85.95±5.36	**91.99±2.36**
3	**98.88±1.70**	40.34±27.07	92.41±4.89	93.73±7.97	95.61±2.39	87.63±9.37	82.86±1.25	87.80±6.16	86.78±6.08	89.00±5.74	97.53±1.16	97.06±1.12	95.61±1.81	90.45±3.44
4	99.90±0.08	99.95±0.10	99.89±0.13	99.89±0.09	99.98±0.03	99.98±0.02	**100.00±0.00**	**100.00±0.00**	99.70±0.20	99.97±0.03	99.88±0.24	99.93±0.05	99.92±0.07	**100.00±0.00**
5	99.43±0.37	99.47±0.79	99.98±0.02	99.96±0.07	**100.00±0.00**	99.68±0.44	99.98±0.02	99.99±0.01	99.97±0.03	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.99±0.01	99.98±0.02
6	89.38±2.08	93.56±3.47	93.53±3.47	96.98±0.91	**97.21±1.36**	87.79±7.48	92.65±2.28	92.64±2.69	93.62±4.14	91.83±2.07	96.20±1.75	94.54±2.47	94.66±3.43	95.05±1.88
OA(%)	97.76±0.27	97.11±0.82	97.66±0.41	97.51±0.40	97.93±0.22	97.50±0.95	98.08±0.09	98.02±0.25	97.42±0.77	97.18±0.83	97.80±0.50	97.21±1.34	97.94±0.30	**98.42±0.21**
AA(%)	**96.5±0.45**	87.10±4.74	95.09±1.37	94.96±1.68	95.89±0.35	94.33±2.47	94.42±0.11	95.08±0.91	93.99±0.89	93.81±1.22	96.00±0.94	94.89±2.30	95.95±0.56	96.08±0.54
$\kappa \times 100$	97.01±0.36	96.13±1.10	96.88±0.55	96.68±0.54	97.24±0.29	96.65±1.27	95.44±2.88	97.36±0.34	96.55±1.03	96.24±1.11	97.06±0.67	96.27±1.79	97.24±0.40	**97.89±0.28**
Training Time(s)	**15.63±1.48**	22.46±0.52	39.24±1.19	18.25±0.39	44.79±2.13	21.98±0.39	42.68±1.45	123.26±5.35	18.03±0.94	108.26±3.53	59.16±5.36	49.26±2.29	130.26±6.26	32.13±1.90
Test Time(s)	2.45±0.35	4.13±0.19	10.88±0.59	3.69±0.08	14.57±1.09	3.72±0.29	**2.08±0.13**	62.32±2.79	3.22±0.11	15.69±1.98	7.26±1.26	5.26±1.02	18.23±2.30	3.65±0.23
Parameters(K)	241.54	46.53	95.54	299.9	105.19	92.71	132.35	106.18	**10.21**	57.21	157.36	185.2	1942.26	12.9
FLOPs(M)	626.79	3.7	69.29	23.59	9.15	4.13	200.3	130.22	7.12	9.52	9.68	19.63	54.26	**0.88**

two modules increases the model's OA by 1.85%. Based on SSC, a comparison between rows 5 and 8 indicates that using CS^{2}CA improves the model's OA by 0.24%. Finally, by comparing rows 4 and 8, we can see that the introduction of the OC-Mamba block boosts the model's OA by 1.95%. This experiment demonstrates the positive impact of the proposed modules on the classification performance of HSI.

G. Classification Comparison With State-of-the-Art Models

To further highlight the advantages of our proposed SliMamba, we conducted a comparative evaluation against state-of-the-art methods that encompass CNN, Transformer, and SSM architectures. Specifically, CVSSN [35], MSCA-DC [63], and HKNAS [44] represent CNN-based methods, while SSFTT [20], MorphFormer [64], and CMT [36] are Transformer-based approaches. MambaHSI [28], MiM [21], 3DSS-Mamba [22], IGSS-Mamba [56], HyperMamba [58], MambaLG [57], and SpiralMamba [59] are categorized as SSM-based methods. Notably, CVSSN and CMT are methods that explicitly incorporate the prior knowledge of the central pixel. Both MiM and SpiralMamba introduce scanning mechanisms centered around the central pixel to enhance local feature aggregation. In contrast, MambaHSI performs image-level classification, whereas the other approaches operate at the patch level. It is also worth mentioning that MSCA-DC, CMT, and 3DSS-Mamba are relatively lightweight models in terms of parameter count and computational complexity.

Tables VIII–XI summarize the OA, AA, κ, training time, testing time, parameters, FLOPs, and per-class accuracy of each method on the Houston, Trento, WHU-Hi-HongHu, and Fanglu Tea Farm datasets. The first and second highest results are highlighted in bold and underlined, respectively. Clearly, our proposed SliMamba outperforms all other methods, achieving the highest OA, AA, and κ (except for AA in Trento and both AA and κ in Fanglu Tea Farm).

In terms of classification accuracy, our model demonstrates consistently superior performance across all datasets. On the Houston dataset, our model achieves an OA of 98.94% $\pm$ 0.23%, outperforming the second-best method by 0.51%. Additionally, our AA and κ scores surpass the second best by 0.37% and 0.55, respectively. On the Trento dataset, which is characterized by a small number of classes and imbalanced sample distributions, our model exceeds the second-best OA by 0.34%. For the WHU-Hi-HongHu dataset, our model delivers outstanding results, attaining the highest accuracy for the vast majority of classes, with improvements of 0.31% in OA, 0.64% in AA, and 0.4 in κ score compared to the second-best approach. Finally, on the Fanglu Tea Farm dataset, although our AA and κ scores are slightly lower than those of the MambaHSI model, our OA is higher by 1.67% and exceeds the second-best method by 0.59%. Moreover, across all datasets, our model consistently achieves either the best or second-best single-class accuracy in most categories.

To further illustrate the classification accuracy of each model, we present the classification maps in Figs. 8–11,

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TABLE X

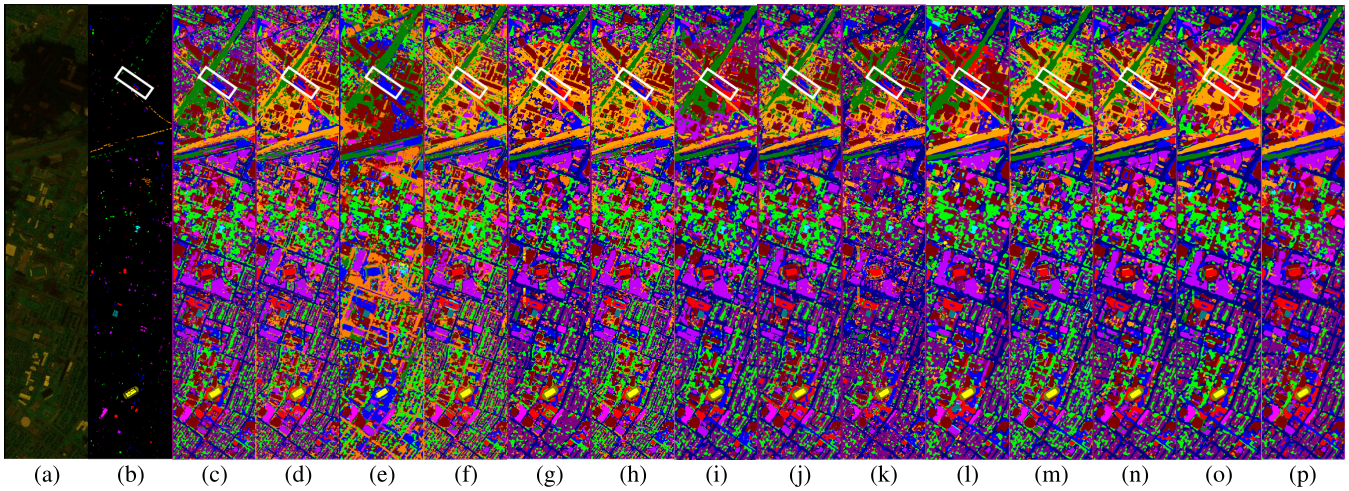
OAs[%], AAs[%], $\kappa \times 100$, TRAINING TIME, AND TESTING TIME STATISTIC FOR THE WHU-HI-HONGHU SCENE

	CNN-based			Transformer-Based			SSM-Based							
Class	CVSSN TCSVT2022	MSCA-DC GRSL2023	HKNAS3D TNNLS2023	SSFTT TGRS2022	MorphFormer TGRS2023	CMT TGRS2024	MambaHSI TGRS2024	MiM IJON2024	3DSS-Mamba TGRS2024	IGSS-Mamba TGRS2024	HyperMamba TGRS2024	MambaLG TGRS2024	SpiralMamba TGRS2025	SliMamba Proposed
1	94.26±2.08	94.06±4.55	96.08±0.21	96.55±1.33	96.50±1.26	94.94±2.07	**98.35±0.65**	96.20±0.97	96.35±1.28	97.49±0.93	97.54±1.01	97.29±0.81	97.14±1.60	97.89±0.60
2	21.61±15.91	14.62±29.24	79.90±6.72	91.62±5.26	91.45±4.80	70.57±19.14	70.44±11.39	70.66±6.92	92.64±2.98	87.63±5.54	**92.68±5.87**	87.80±8.41	91.77±4.45	84.24±3.38
3	96.40±2.72	95.20±2.04	95.64±0.70	95.76±1.55	96.42±2.07	94.77±1.31	96.27±1.39	92.78±3.06	96.00±1.03	**96.53±1.78**	95.65±3.69	86.10±11.56	93.17±8.41	95.78±1.25
4	99.72±0.16	99.71±0.22	99.79±0.07	99.44±0.15	99.29±0.17	97.71±1.56	**99.93±0.05**	99.69±0.09	99.44±0.20	99.66±0.12	99.70±0.20	99.75±0.08	99.60±0.18	99.75±0.05
5	62.16±12.89	27.18±27.32	73.68±6.48	85.98±5.39	85.99±5.49	70.53±6.70	39.18±28.71	84.38±3.19	82.26±6.53	**91.00±5.29**	72.34±8.79	90.13±6.49	85.60±12.65	88.78±0.62
6	97.00±1.49	94.65±1.97	98.34±0.27	98.72±0.36	98.43±0.75	96.58±1.71	94.76±2.41	97.43±0.99	98.41±0.65	97.88±1.00	**99.48±0.10**	98.50±1.06	98.83±0.32	98.97±0.21
7	90.80±3.68	86.95±5.21	87.99±3.58	92.44±2.04	89.63±5.08	83.27±6.37	95.51±1.64	92.88±1.07	93.70±0.80	**95.72±0.92**	94.29±2.99	94.31±1.94	93.47±2.49	94.13±0.84
8	0.00±0.00	0.00±0.00	36.10±8.23	67.33±7.04	69.74±7.72	69.09±10.39	1.22±2.33	42.44±8.38	71.71±11.05	81.40±8.29	73.65±8.61	67.72±9.48	74.55±5.52	**83.96±1.14**
9	93.07±3.42	87.63±4.54	97.08±1.33	98.87±0.55	97.77±1.25	97.94±0.76	96.25±1.29	97.46±1.67	98.55±0.38	98.55±1.05	98.96±0.62	**99.45±0.33**	98.79±0.72	99.31±0.32
10	81.46±6.29	81.37±8.24	86.85±1.94	91.43±3.98	93.27±1.36	80.88±11.98	79.29±4.35	87.72±2.51	86.19±2.43	93.68±3.08	**95.35±2.87**	93.82±4.17	92.87±3.03	94.60±2.05
11	70.23±7.21	64.35±5.35	77.32±3.29	88.40±4.51	89.64±2.17	83.32±5.16	64.74±4.80	79.13±4.95	89.93±1.76	92.16±2.25	**95.44±1.94**	88.83±6.57	91.05±4.48	93.89±1.69
12	61.41±3.79	61.31±6.52	68.64±3.21	83.64±4.99	85.48±2.62	50.59±14.95	61.57±3.03	71.55±3.54	81.14±2.74	83.28±2.50	81.93±14.60	84.53±6.17	81.14±5.35	**87.15±1.12**
13	84.56±2.75	81.41±6.19	85.28±2.00	89.13±4.06	91.39±2.56	74.65±15.45	84.84±3.27	83.06±0.78	92.43±1.00	92.71±1.52	90.97±1.74	92.14±3.60	91.49±1.93	**93.12±0.63**
14	74.18±7.56	72.90±8.32	90.66±2.21	94.58±2.40	93.72±1.52	75.08±23.22	69.15±5.77	85.04±4.96	94.37±1.94	91.70±2.18	**96.44±1.40**	89.67±5.68	93.04±3.03	95.29±1.06
15	0.00±0.00	0.00±0.00	43.71±28.91	76.37±12.15	85.80±3.33	91.44±3.86	0.00±0.00	36.93±23.05	85.92±2.80	88.06±4.72	87.94±8.52	79.57±5.38	83.00±4.92	**91.92±3.68**
16	**99.17±0.58**	98.59±0.63	96.22±1.22	96.39±2.00	96.32±2.11	93.95±5.18	96.03±3.07	92.05±3.19	96.75±1.87	96.40±1.13	98.50±0.87	98.25±1.34	97.56±0.77	97.81±0.73
17	0.00±0.00	0.26±0.51	83.72±5.05	92.17±3.25	93.81±2.70	66.93±30.68	46.79±25.51	58.06±27.78	92.15±2.52	93.93±2.74	75.76±17.88	68.42±29.27	91.30±6.49	**93.96±1.72**
18	0.00±0.00	5.74±11.48	81.02±8.30	90.09±5.04	87.90±5.39	84.53±14.16	2.24±3.23	70.55±13.93	87.30±3.53	**96.68±0.71**	94.97±2.36	87.08±8.60	92.76±4.57	94.88±2.09
19	73.42±8.01	67.08±10.97	90.62±2.11	93.42±1.36	94.03±1.39	92.78±3.61	81.54±5.49	90.02±2.28	91.96±3.59	93.89±0.74	95.51±0.57	**95.82±1.48**	93.46±1.71	94.20±0.81
20	0.00±0.00	2.53±5.06	89.27±2.93	90.64±2.95	90.93±2.04	79.41±6.71	3.87±3.35	78.42±5.48	92.23±3.27	91.97±4.45	94.00±4.40	**95.79±0.37**	95.42±1.36	91.88±1.96
21	0.00±0.00	0.00±0.00	0.00±0.00	58.26±25.52	72.47±11.79	68.39±11.22	0.00±0.00	4.80±6.14	57.72±15.53	77.57±11.46	47.51±16.15	62.45±23.40	68.16±20.36	**80.82±3.22**
22	13.32±20.44	11.22±22.44	77.59±3.79	81.48±8.44	85.59±5.60	86.03±6.02	11.92±14.01	81.09±2.83	83.64±8.05	93.79±2.71	**94.59±2.44**	89.88±3.44	92.07±3.08	93.46±2.71
OA(%)	87.63±0.63	85.78±0.97	93.01±0.50	95.54±0.47	95.68±0.40	90.69±2.39	88.29±1.07	92.65±0.72	95.60±0.18	96.65±0.36	96.33±0.38	95.53±0.76	96.05±0.23	**96.96±0.19**
AA(%)	55.13±1.44	52.12±4.33	78.89±2.25	88.76±2.03	90.25±1.05	81.97±4.03	58.81±3.40	76.93±1.62	89.13±1.37	92.35±1.02	89.69±1.65	88.51±3.60	90.74±0.95	**92.99±0.46**
$\kappa \times 100$	84.11±0.86	81.69±1.29	91.13±0.64	94.36±0.60	94.53±0.51	88.25±3.01	84.40±1.42	90.68±0.92	94.43±0.23	95.76±0.45	95.36±0.48	94.34±0.96	95.01±0.29	**96.16±0.24**
Training Time(s)	366.35±22.12	246.83±4.69	379.16±9.32	244.42±3.16	526.98±10.44	**138.29±2.19**	180.26±4.31	350.35±3.51	182.23±3.45	289.26±4.13	243.56±3.77	230.02±2.52	332.26±6.26	220.23±2.97
Test Time(s)	21.26±1.78	67.15±1.06	145.47±4.46	115.61±2.19	143.67±3.57	17.15±0.74	**15.42±1.16**	201.28±2.59	33.23±0.88	79.26±3.2	38.26±2.01	30.26±1.26	89.32±5.32	23.23±2.76
Parameters(K)	270.91	145.93	112.9	1254.8	256	94.77	392.06	107.22	**10.74**	57.74	162.11	196.26	1952.24	15.1
FLOPs(M)	698.71	11.54	119	102.68	32.69	4.13	450.62	130.22	7.12	9.52	9.76	21.35	57.11	**1.01**

TABLE XI

OAs[%], AAs[%], $\kappa \times 100$, TRAINING TIME, AND TESTING TIME STATISTIC FOR THE FANGLU TEA FARM SCENE

	CNN-based			Transformer-Based			SSM-Based							
Class	CVSSN TCSVT2022	MSCA-DC GRSL2023	HKNAS3D TNNLS2023	SSFTT TGRS2022	MorphFormer TGRS2023	CMT TGRS2024	MambaHSI TGRS2024	MiM IJON2024	3DSS-Mamba TGRS2024	IGSS-Mamba TGRS2024	HyperMamba TGRS2024	MambaLG TGRS2024	SpiralMamba TGRS2025	SliMamba Proposed
1	92.97±2.14	96.61±2.69	99.33±0.48	99.35±0.50	99.36±0.67	68.62±20.10	99.36±0.76	99.71±0.14	97.47±1.66	98.94±1.12	99.15±0.60	98.95±0.50	98.56±1.54	**99.83±0.08**
2	90.68±4.82	51.59±32.57	88.98±2.65	80.92±7.41	81.50±4.70	72.41±12.38	**99.68±0.33**	91.81±5.71	89.82±2.28	88.94±3.60	94.09±3.95	87.52±5.42	90.43±3.67	95.73±2.35
3	97.73±0.85	98.33±0.81	99.11±0.31	99.19±0.53	98.86±0.53	85.22±17.11	95.66±2.13	99.29±0.44	98.56±0.49	99.23±0.83	98.82±0.74	99.00±0.54	99.21±0.70	**99.50±0.22**
4	83.26±9.22	0.00±0.00	94.26±4.07	73.21±16.11	82.26±14.05	64.78±9.77	**100.00±0.00**	73.05±32.35	85.45±11.56	52.73±20.76	94.45±9.51	2.68±5.36	97.61±2.44	92.63±12.83
5	98.95±0.74	98.84±0.65	99.79±0.24	99.40±0.85	99.64±0.24	99.67±0.55	99.89±0.06	**99.99±0.01**	99.40±0.30	99.62±0.32	99.95±0.05	99.90±0.11	99.77±0.32	99.96±0.07
6	92.45±6.82	72.72±4.70	**98.57±1.26**	92.82±8.16	95.12±1.07	93.53±2.63	97.86±1.25	80.22±11.25	87.15±2.83	83.54±5.91	96.21±2.11	94.13±4.32	88.66±7.63	90.64±1.26
7	81.41±9.38	39.00±23.88	92.63±5.37	91.37±7.42	97.22±3.92	90.57±12.30	**99.95±0.10**	91.92±11.29	93.49±4.62	98.80±1.32	97.08±1.33	89.67±15.58	98.13±2.63	98.76±2.49
8	79.69±4.70	55.19±8.67	90.58±3.01	81.04±6.17	83.65±3.81	66.91±12.70	**98.67±0.89**	85.33±9.69	75.80±8.91	68.88±6.64	84.85±12.43	83.10±4.17	84.71±9.71	92.97±2.55
9	99.63±0.32	99.87±0.11	99.99±0.01	99.85±0.13	99.56±0.47	99.54±0.53	99.21±0.49	99.76±0.48	99.87±0.08	99.75±0.47	99.93±0.13	99.98±0.02	99.91±0.12	**100.00±0.00**
10	76.09±13.34	97.18±1.15	98.85±0.99	98.49±1.02	98.91±1.14	92.55±5.15	**99.75±0.18**	97.54±2.80	99.39±0.38	92.64±8.51	99.21±0.42	98.22±2.11	98.60±1.28	99.50±0.48
OA(%)	96.04±0.77	93.60±1.36	98.50±0.19	97.62±0.16	97.68±0.26	85.92±9.16	97.42±1.15	98.19±0.80	97.28±0.25	97.27±0.53	98.37±0.40	97.63±0.39	98.22±0.46	**99.09±0.16**
AA(%)	89.29±2.53	70.93±3.75	96.21±0.64	91.56±1.29	93.61±1.06	83.38±1.10	**99.00±0.41**	91.86±6.75	92.64±1.13	88.31±3.71	96.37±2.07	85.31±2.06	95.56±1.10	96.95±1.27
$\kappa \times 100$	94.11±1.17	90.30±2.18	97.79±0.28	96.46±0.25	96.57±0.38	80.81±11.37	**98.67±1.63**	97.31±1.20	95.97±0.37	95.93±0.80	97.59±0.60	96.48±0.58	97.36±0.68	98.65±0.24
Training Time(s)	36.35±2.12	24.23±1.23	58.53±2.35	**21.39±0.99**	36.78±1.56	39.45±2.12	114.89±4.78	234.67±9.87	47.89±1.98	189.26±6.59	86.32±4.26	59.26±4.26	198.26±6.26	41.23±1.67
Test Time(s)	**5.26±0.78**	10.83±0.46	21.49±0.94	6.46±0.45	22.78±0.89	16.34±0.78	37.95±1.67	101.89±4.32	27.65±1.23	59.26±2.26	46.26±1.25	29.26±2.23	98.48±5.12	11.34±0.67
Parameters(K)	244.27	16.89	97.66	378.5	103.21	93.23	250.36	106.44	**10.35**	57.35	158.26	187.34	1948.43	13.16
FLOPs(M)	632.63	1.36	71.57	30.1	9.91	4.13	284.06	130.22	7.12	9.52	9.7	20.16	55.23	**0.89**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l) (m) (n) (o) (p)

Fig. 8. False color, ground truth map of the Houston dataset, and classification maps generated by different models (corresponding to the single experiment in Table VIII). (a) False color. (b) Ground truth. (c) CVSSN. (d) MSCA-DC. (e) HKNAS3D. (f) SSFTT. (g) MorphFormer. (h) CMT. (i) MambaHSI. (j) MiM. (k) 3DSS-Mamba. (l) IGSS-Mamba. (m) HyperMamba. (n) MambaLG. (o) SpiralMamba. (p) SliMamba.

corresponding to the highest OA achieved in ten trials. As shown, the maps generated by our SliMamba generally contain less noise than those produced by other methods across the four benchmark datasets, and closely match the ground

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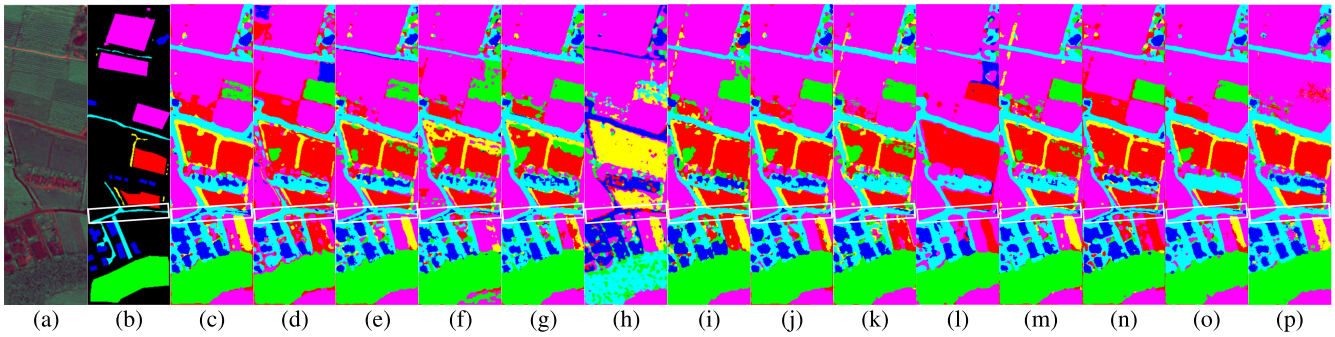


Fig. 9. False color, ground truth map of the Trento dataset, and classification maps generated by different models (corresponding to the single experiment in Table IX). (a) False color. (b) Ground truth. (c) CVSSN. (d) MSCA-DC. (e) HKNAS3D. (f) SSFTT. (g) MorphFormer. (h) CMT. (i) MambaHSI. (j) MiM. (k) 3DSS-Mamba. (l) IGSS-Mamba. (m) HyperMamba. (n) MambaLG. (o) SpiralMamba. (p) SliMamba.

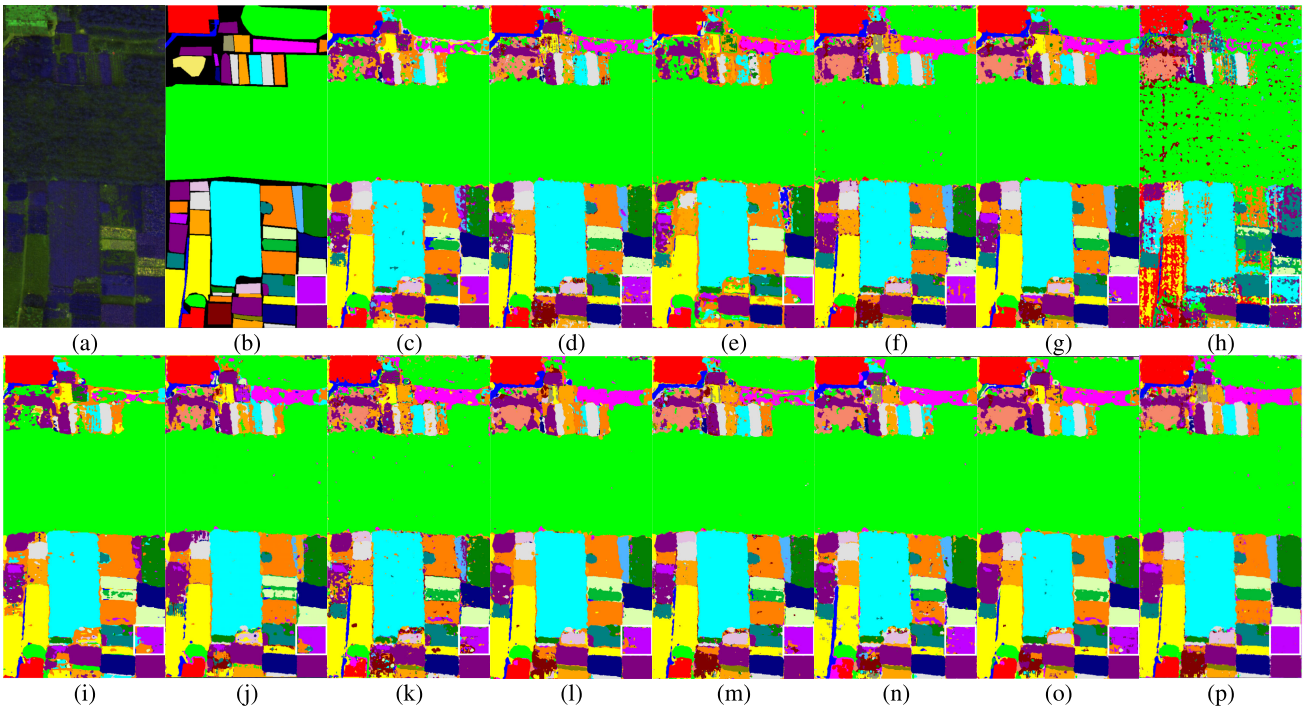


Fig. 10. False color, ground truth map of the WHU-Hi-HongHu dataset, and classification maps generated by different models (corresponding to the single experiment in Table X). (a) False color. (b) Ground truth. (c) CVSSN. (d) MSCA-DC. (e) HKNAS3D. (f) SSFTT. (g) MorphFormer. (h) CMT. (i) MambaHSI. (j) MiM. (k) 3DSS-Mamba. (l) IGSS-Mamba. (m) HyperMamba. (n) MambaLG. (o) SpiralMamba. (p) SliMamba.

truth. Specifically, regions of interest are highlighted with white boxes to facilitate detailed comparison of different models. In the Houston dataset, the white box covers three ground-truth land cover types: highway (dark green), healthy grass (red), and stressed grass (blue). Compared with other algorithms, our method can clearly distinguish these three classes and extend the correct predictions to neighboring regions based on the ground truth. In the Trento dataset, the white box marks a region labeled as roads (cyan). Other methods tend to misclassify this region with the building class (blue), whereas our approach almost never introduces such building noise. In the WHU-Hi-HongHu dataset, only our method can almost perfectly identify Brassica Chinensis within the white box. A similar trend is observed on the Fanglu Tea Farm dataset.

However, in terms of training time and testing time, our SliMamba does not outperform the SOTA methods. This may be attributed to the multiple convolutional operations employed in the SSC module, the complex operations in the OC-Mamba module, all of which contribute to less efficient time performance. Nevertheless, for testing time, the model consistently ranks among the top 3 or 4 methods, and even achieves a top 2 position in Houston dataset. Fortunately, our model demonstrates clear advantages in terms of both parameters and FLOPs. Across all four datasets, its parameter count is comparable to that of the most lightweight 3DSS-Mamba, while its FLOPs are reduced to approximately one seventh of that model, resulting in the lowest computational cost among all compared approaches.

In summary, our model not only achieves the best classification performance but also maintains a highly lightweight structure, effectively balancing model complexity and predictive accuracy.

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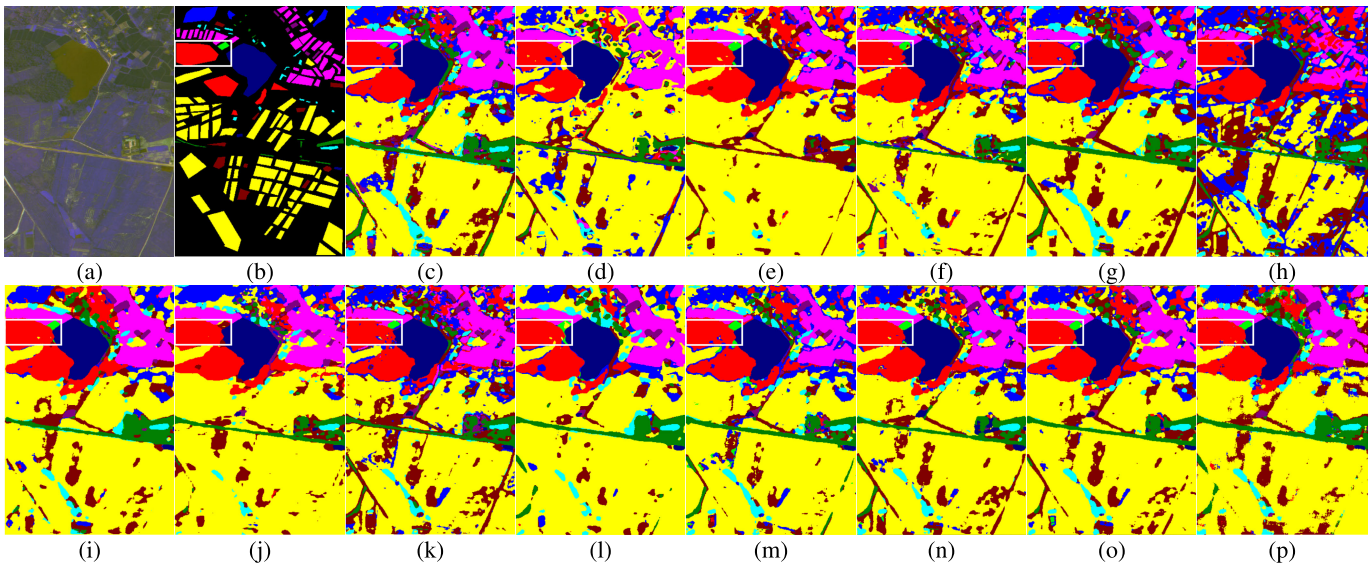


Fig. 11. False color, ground truth map of the Fanglu Tea Farm dataset, and classification maps generated by different models (corresponding to the single experiment in Table XI). (a) False color. (b) Ground truth. (c) CVSSN. (d) MSCA-DC. (e) HKNAS3D. (f) SSFTT. (g) MorphFormer. (h) CMT. (i) MambaHSI. (j) MiM. (k) 3DSS-Mamba. (l) IGSS-Mamba. (m) HyperMamba. (n) MambaLG. (o) SpiralMamba. (p) SliMamba.

V. Conclusion and Future Research

In this article, we developed a novel lightweight network model named SliMamba for advanced land cover classification of hyperspectral remote sensing images. The framework adopts a dual-branch architecture to model both dimension-reduced and original spectral features. Thanks to the integration of SSC and OC-Mamba, the model is capable of capturing cross-channel features as well as global–local spectral–spatial features based on center pixel priors. Extensive experiments demonstrate that SliMamba efficiently overcomes the limitations of existing SSM-based hyperspectral imaging architectures in terms of both performance and scalability, providing a practical solution for HSIC tasks. Future research will focus on simplifying the complex convolutional operations and Mamba scanning mechanisms to further improve time efficiency, as well as extending the application of SliMamba to more challenging hyperspectral scenarios.

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References

[1] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[2] N. Wambugu et al., "Hyperspectral image classification on insufficient-sample and feature learning using deep neural networks: A review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102603.

[3] M. R. Ahmed et al., "Multiclass classification on soybean and weed species using a customized greenhouse robotic and hyperspectral combination system," *J. ASABE*, vol. 65, no. 5, pp. 1071–1080, 2022.

[4] H. Zhang et al., "Hyperspectral image classification on large-scale agricultural crops: The Heilongjiang benchmark dataset, validation procedure, and baseline results," *Remote Sens.*, vol. 16, no. 3, p. 478, Jan. 2024.

[5] Y. Yang et al., "Fine-scale mangrove species classification based on UAV multispectral and hyperspectral remote sensing using machine learning," *Remote Sens.*, vol. 16, no. 16, p. 3093, Aug. 2024.

[6] S.-F. Huang, S.-C. Lu, A. Mukundan, R. Karmakar, and H.-C. Wang, "Application of hyperspectral imaging in environmental monitoring: Air pollution classification and detection," *Opt. Exp.*, vol. 33, no. 8, pp. 17955–17964, 2025.

[7] N. Li, S. Jiang, J. Xue, S. Ye, and S. Jia, "Texture-aware self-attention model for hyperspectral tree species classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502215.

[8] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[9] F. Zhang, B. Du, L. Zhang, and L. Zhang, "Hierarchical feature learning with dropout k-means for hyperspectral image classification," *Neurocomputing*, vol. 187, pp. 75–82, Apr. 2016.

[10] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[11] L. Ma, M. M. Crawford, and J. Tian, "Local manifold learning-based *k*-nearest-neighbor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4099–4109, Nov. 2010.

[12] S. Yu, S. Jia, and C. Xu, "Convolutional neural networks for hyperspectral image classification," *Neurocomputing*, vol. 219, pp. 88–98, Jan. 2017.

[13] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "A new deep convolutional neural network for fast hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 120–147, Nov. 2018.

[14] J. Li, X. Zhao, Y. Li, Q. Du, B. Xi, and J. Hu, "Classification of hyperspectral imagery using a new fully convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 292–296, Feb. 2018.

[15] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[16] S. Jia, S. Jiang, S. Zhang, M. Xu, and X. Jia, "Graph-in-graph convolutional network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1157–1171, Jan. 2024.

[17] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[18] X. Zhang, Y. Sun, K. Jiang, C. Li, L. Jiao, and H. Zhou, "Spatial sequential recurrent neural network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4141–4155, Nov. 2018.

[19] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

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[20] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522214.

[21] W. Zhou, S.-I. Kamata, H. Wang, M. S. Wong, and H. C. Hou, "Mamba-in-Mamba: Centralized Mamba-cross-scan in tokenized Mamba model for hyperspectral image classification," *Neurocomputing*, vol. 613, Jan. 2025, Art. no. 128751.

[22] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-Mamba: 3D-spectral–spatial Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534216.

[23] J. Yao, D. Hong, C. Li, and J. Chanussot, "SpectralMamba: Efficient Mamba for hyperspectral image classification," 2024, *arXiv:2404.08489*.

[24] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[25] T. Lu, M. Liu, W. Fu, and X. Kang, "Grouped multi-attention network for hyperspectral image spectral–spatial classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507912.

[26] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.

[27] C. Rodarmel and J. Shan, "Principal component analysis for hyperspectral image classification," *Surveying Land Inf. Sci.*, vol. 62, no. 2, pp. 115–122, 2002.

[28] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[29] R. Reshma, V. Sowmya, and K. P. Soman, "Retraction note: Effect of Legendre–Fenchel denoising and SVD-based dimensionality reduction algorithm on hyperspectral image classification," *Neural Comput. Appl.*, vol. 36, no. 19, p. 11695, Jul. 2024.

[30] C. Liu, J. Li, L. He, A. Plaza, S. Li, and B. Li, "Naive Gabor networks for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 376–390, Jan. 2021.

[31] S. Jia et al., "3-D Gabor convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[32] Z. Zhao, X. Xu, J. Li, S. Li, and A. Plaza, "Gabor-modulated grouped separable convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518817.

[33] T. V. N. Prabhakar and P. Geetha, "Two-dimensional empirical wavelet transform based supervised hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 133, pp. 37–45, Nov. 2017.

[34] M. Ahmad, U. Ghous, M. Usama, and M. Mazzara, "WaveFormer: Spectral–spatial wavelet transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[35] M. Li, Y. Liu, G. Xue, Y. Huang, and G. Yang, "Exploring the relationship between center and neighborhoods: Central vector oriented self-similarity network for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1979–1993, Apr. 2023.

[36] S. Jia, Y. Wang, S. Jiang, and R. He, "A center-masked transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510416.

[37] H. Lee and H. Kwon, "Contextual deep CNN based hyperspectral classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 3322–3325.

[38] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[39] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[40] C. Yu, R. Han, M. Song, C. Liu, and C.-I. Chang, "A simplified 2D-3D CNN architecture for hyperspectral image classification based on spatial–spectral fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2485–2501, 2020.

[41] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[42] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. J. Plaza, and F. Pla, "Deep pyramidal residual networks for spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Feb. 2019.

[43] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[44] D. Wang, B. Du, L. Zhang, and D. Tao, "HKNAS: Classification of hyperspectral imagery based on hyper kernel neural architecture search," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 13631–13645, Oct. 2024.

[45] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[46] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[47] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[48] D. Y. Fu, T. Dao, K. K. Saab, A. W. Thomas, A. Rudra, and C. Ré, "Hungry hungry hippos: Towards language modeling with state space models," 2022, *arXiv:2212.14052*.

[49] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

[50] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.

[51] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision Mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[52] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[53] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-Mamba for large remote sensing image dense prediction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5633314.

[54] E. Zhu, Z. Chen, D. Wang, H. Shi, X. Liu, and L. Wang, "UNetMamba: An efficient UNet-like Mamba for semantic segmentation of high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[55] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into Mamba for remote sensing image binary change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4405016.

[56] Y. He, B. Tu, P. Jiang, B. Liu, J. Li, and A. Plaza, "IGroupSS-Mamba: Interval group spatial–spectral Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538817.

[57] Z. Pan, C. Li, A. Plaza, J. Chanussot, and D. Hong, "Hyperspectral image classification with Mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 5602814.

[58] Q. Liu, J. Yue, Y. Fang, S. Xia, and L. Fang, "HyperMamba: A spectral–spatial adaptive Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5536514.

[59] X. Tang et al., "SpiralMamba: Spatial–spectral complementary Mamba with spatial spiral scan for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5510319.

[60] M. Ahmad, M. Usama, M. Mazzara, and S. Distefano, "WaveMamba: Spatial–spectral wavelet Mamba for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[61] Y. Xu et al., "Spatial–Spectral 1DSwin transformer with groupwise feature tokenization for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5516616.

[62] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-Hi: UAV-borne hyperspectral with high spatial resolution (H^2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.

[63] Z. Ye, C. Li, Q. Liu, L. Bai, and J. E. Fowler, "Computationally lightweight hyperspectral image classification using a multiscale depth-wise convolutional network with channel attention," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[64] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.

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